

M.TECH THESIS

**Secure IoT Data Analytics via q-Sigmoid Enhanced
Federated Learning for Threat Detection**

by

Name

Shalu Tiwari

SAU/CS/MTech(CS)/2024/07

Under the Supervision of

Dr. Shachi Sharma

Submitted in partial fulfillment of the requirements for the award of the degree of

Master of Technology (M.Tech) in Computer Science and Engineering

to the



Department of Computer Science and Engineering

Faculty of Engineering and Technology

South Asian University, New Delhi – 110068, India

May, 2026

Certificate

This is to certify that the thesis entitled “**Secure IoT Data Analytics via q-Sigmoid Enhanced Federated Learning for Threat Detection**” submitted by **Shalu Tiwari** to the Department of Computer Science and Engineering, Faculty of Engineering and Technology, South Asian University, New Delhi in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Computer Science and Engineering**, is a record of the bonafide research work carried out by her under my supervision and guidance.

The results contained in this thesis have not been submitted in part or full to any other university or institute for the award of any degree or diploma.

Dr. Shachi Sharma

(Supervisor)

Department of Computer Science

South Asian University

New Delhi

Declaration

I hereby declare that the thesis entitled “**Secure IoT Data Analytics via q-Sigmoid Enhanced Federated Learning for Threat Detection,**” being submitted to the Department of Computer Science and Engineering, Faculty of Engineering and Technology, South Asian University, New Delhi in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Computer Science and Engineering** contains the original work carried out by me under the supervision of Dr. Shachi Sharma. The research work reported in this thesis is original and has not been submitted either in part or full to any university or institution for the award of any degree or diploma.

Shalu Tiwari

Enrollment No.:

SAU/CS/MTech(CS)/2024/07

Acknowledgment

First of all, I would like to express my sincere gratitude to my supervisor, Dr. Shachi Sharma, for her continuous support of this research work, for her patience, motivation, and her immense knowledge. Her way of guidance always encouraged me to think in-depth about my research topic. Her precious ideas, not only limited to my research topic, have helped me and guided me in different aspects of life. I want to admire all her contributions of time and ideas that made my research work more productive. I have to express my heartfelt thanks to my family my mother, brothers, and sisters who have always provided me unconditional support and love throughout my years of study. Finally, I would like to thank all my friends who helped me and motivated me in difficult times.

Shalu Tiwari

SAU/CS/MTech(CS)2024/07

May 2026

Abstract

IoT devices are now everywhere — from hospitals and factories to smart homes and city infrastructure. While this connectivity brings many benefits, it also opens the door to serious cyber threats like DDoS attacks, botnet infections, and unauthorized access. Detecting these threats is not straightforward, especially when devices are spread across different locations and generate very different kinds of traffic data. Traditional intrusion detection systems generally assume that all data can be collected and processed at a central server. In IoT environments, however, this assumption is difficult to satisfy because transmitting large volumes of sensitive device data raises both privacy and scalability concerns.

Federated Learning (FL) addresses this issue by allowing devices to train models locally while sharing only model parameters with the server. Although this framework improves privacy preservation, practical IoT deployments introduce another challenge: the data distribution across devices is highly heterogeneous. Some clients may observe mostly normal traffic, whereas others may experience a significantly larger number of attack samples. Under such imbalanced conditions, conventional loss functions such as Binary Cross-Entropy (BCE) often fail to learn robust decision boundaries.

To address this limitation, this thesis investigates the use of a Tsallis-statistics-based loss function known as q -Sigmoid loss. The q parameter controls the sensitivity of the loss function and can improve robustness against noisy and skewed data distributions. Building upon this idea, a PerClient- q -Sigmoid strategy is proposed in which each client dynamically selects its own q value according to the imbalance level of its local dataset. Clients with highly skewed attack distributions are assigned larger q values, while clients with relatively balanced data operate closer to standard sigmoid behaviour. Since the modification is performed entirely on the client side, the central aggregation mechanism of Federated Learning remains unchanged.

The proposed framework is evaluated on three publicly available intrusion detection datasets: UNSW-NB15, WUSTL-IIoT-2021, and CSE-CIC-IDS2018. Experiments are conducted under both balanced and non-IID settings to simulate realistic federated environments. The results demonstrate that the proposed method consistently outperforms standard BCE-based approaches across all datasets. The performance gain becomes particularly significant in highly imbalanced scenarios, indicating that client-specific q adaptation improves the robustness of federated intrusion detection systems.

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Chapter 1

Introduction

1.1 Background of the Study

The rapid growth of Internet of Things (IoT) devices has significantly transformed modern digital infrastructure across domains such as healthcare, smart cities, industrial automation, transportation, and intelligent surveillance systems. However, the large-scale deployment of interconnected IoT devices has also introduced major cybersecurity challenges due to device heterogeneity, limited computational resources, weak security configurations, and continuous network communication. These vulnerabilities make IoT environments highly susceptible to cyberattacks such as Distributed Denial of Service (DDoS), botnets, reconnaissance attacks, and unauthorized access. Consequently, efficient and intelligent Intrusion Detection Systems (IDS) have become essential for securing IoT networks against evolving cyber threats.. Industry reports indicate that the number of connected IoT devices grew at an annual rate of approximately 14% in 2025, reaching around 39 billion devices by 2030 and exceeding 50 billion by 2035. [1]. IoT adoption spans multiple verticals, including wearable devices, smart vehicles, smart homes, smart cities, and industrial equipment. In particular, industrial sectors are expected to witness significant growth in IoT integration due to the convergence of artificial intelligence, augmented reality, and the large-scale deployment of private 5G networks. These technologies enable the seamless integration of digital twins into existing industrial processes [2]. While IoT enables automation and intelligent decision-making, it also significantly increases the attack surface of networks, exposing them to malicious devices, unauthorized access, and various cyber-security threats [3], [4]. such as denial-of-service , man-in-the-middle attacks,and botnet recruiting [5] and AI-Driven Cybersecurity Threats like Deepfakes & Synthetic Media, Adversarial AI Attacks ,Automated Malware Generation,AI-Powered Phishing & Scams ,Social Engineering Automation [6]

Ensuring secure network management in large-scale IoT environments is therefore a critical challenge, as traditional centralized monitoring solutions often struggle with scalability, latency, and privacy concerns [7], [8]. Security in IoT networks depends heavily on the ability to identify and monitor connected devices. Proper device identification helps network administrators recognize unauthorized or suspicious devices, restrict illegal access, and reduce the risk of cyberattacks. To solve these problems, researchers have

explored several machine learning techniques such as device fingerprinting, traffic analysis, and anomaly detection. Although these approaches can achieve good detection accuracy, most of them rely on collecting data at a centralized server, which creates privacy and scalability concerns in large IoT environments.

Federated Learning (FL) has emerged as an alternative approach that allows IoT devices to collaboratively train machine learning models without sending raw data to a central location [9]. Instead of sharing private traffic information, devices exchange only model parameters or updates. This not only improves privacy preservation but also reduces communication overhead across the network. Recent research has demonstrated the usefulness of FL for applications such as intrusion detection, anomaly detection, and intelligent network management in IoT systems [10].

However, deploying FL in practical IoT environments is still challenging. IoT devices often generate highly heterogeneous traffic patterns, and the data available at different clients may be noisy or severely imbalanced. In addition, federated systems remain vulnerable to unreliable updates and adversarial behavior, which can negatively affect detection performance. Traditional loss functions, including cross-entropy and sigmoid-based losses, generally struggle under these conditions because they are sensitive to outliers and skewed class distributions.

To improve robustness, researchers have recently explored Tsallis entropy-based learning methods. In particular, the q-sigmoid loss function introduces a tunable parameter that controls the sensitivity of the learning process and can reduce the influence of noisy samples and extreme data imbalance [11]. Incorporating such robust loss functions into federated learning can improve the stability and effectiveness of intrusion detection systems operating in distributed IoT environments.

Despite recent progress, only limited work has investigated the integration of q-sigmoid-based learning with federated frameworks for IoT security applications. Existing studies also tend to overlook important issues such as client heterogeneity, privacy-aware learning, and reliable detection under highly imbalanced traffic conditions. Motivated by these challenges, this thesis proposes a q-sigmoid enhanced federated learning framework for secure IoT network management. The proposed approach aims to identify malicious devices and network attacks while preserving user privacy and improving robustness against noisy and imbalanced traffic data.

. Extensive experiments on multiple IoT datasets, including UNSW-NB, WUSTL-IIOT-2021 and CSE-CIC-IDS2018. demonstrate that our approach achieves superior detection performance compared to conventional cross-entropy or standard sigmoid-based models.

1.2 IoT Attacks and Vulnerabilities

The rapid growth of the Internet of Things (IoT) has significantly transformed modern communication systems by enabling billions of interconnected devices to exchange information in real time. IoT technologies are now widely deployed in smart homes, healthcare systems, industrial automation, intelligent transportation, agriculture, and military applications. However, the large-scale deployment of heterogeneous and resource-constrained devices has also expanded the cyber-attack surface considerably. IoT devices are highly vulnerable to various security threats including Distributed Denial of Service (DDoS) attacks, botnet attacks, malware injection, spoofing, phishing, ransomware, man-in-the-middle attacks, and unauthorized access. Recently, AI-driven cyber threats such as deepfake-based social engineering, automated malware generation, adversarial attacks, and intelligent phishing campaigns have further increased the complexity of cybersecurity challenges in IoT environments. Many IoT devices are designed with limited processing power, restricted memory, and lightweight security mechanisms. Because of these limitations, they are often unable to withstand modern cyber threats and sophisticated network attacks. As IoT deployments continue to grow, protecting these devices has become an important challenge in network security.

To address this issue, Intrusion Detection Systems (IDS) are widely used as a critical layer of defense in IoT environments. An IDS monitors network traffic and device activity in order to detect suspicious behavior, malicious communication patterns, and unauthorized access attempts. By continuously analyzing system activity in real time, IDS can help identify potential attacks at an early stage and improve the overall security of IoT networks.

1.3 Challenges in Designing IoT IDS

Developing effective intrusion detection systems for IoT networks is difficult because IoT environments differ significantly from traditional computer networks. An IoT ecosystem typically contains a large number of devices with different hardware designs, communication standards, and operational capabilities. These devices often have limited computational resources, including low processing power, small memory capacity, restricted storage, and limited battery life. As a result, many conventional security techniques are too expensive or resource-intensive to be directly deployed in IoT settings.

Another major challenge comes from the nature of IoT traffic itself. Data generated by IoT devices is highly dynamic and varies across devices and applications. In many cases, the data distribution is non-IID, meaning that each device may observe very different traffic patterns and attack behaviors. This heterogeneity makes it difficult for machine learning models to generalize effectively across the entire network.

In addition, IoT systems are usually distributed over large-scale environments, which creates further issues related to scalability, communication cost, and latency. Centralized intrusion detection approaches often require continuous data transfer to a central server, increasing network overhead and raising privacy concerns.

The quality of IoT security datasets also introduces difficulties for model training. Malicious traffic samples are generally much fewer than normal traffic samples, leading to severe class imbalance. Noisy data and incomplete labels can further reduce detection accuracy and produce biased learning behavior. Moreover, IoT networks remain vulnerable to adversarial attacks, poisoning attacks, and privacy leakage during communication between devices and servers.

These limitations demonstrate the need for intrusion detection mechanisms that are not only accurate, but also scalable, privacy-preserving, and robust enough to operate effectively in distributed IoT environments.

1.4 ML/AI/FL-Based IDS Approaches

Traditional rule-based intrusion detection methods are often unable to handle the complexity and continuously evolving nature of cyberattacks in IoT networks. Because of this, researchers have increasingly focused on Machine Learning (ML) and Artificial Intelligence (AI) techniques to improve IoT security. Unlike static rule-based systems, ML-based intrusion detection models can learn patterns directly from network traffic and identify suspicious or abnormal activities more effectively.

Several machine learning and deep learning approaches have been explored for IoT threat detection, including Logistic Regression, Support Vector Machines (SVM), Random Forests, Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Long Short-Term Memory (LSTM) networks. These models have shown promising results in detecting different types of attacks and malicious behavior in network traffic.

Despite their effectiveness, many existing AI-based intrusion detection systems depend on centralized learning architectures where data from multiple devices is collected and processed at a central server. In IoT environments, this approach creates serious concerns related to user privacy, communication overhead, and scalability. Transferring large amounts of sensitive device data across the network may also increase the risk of information leakage.

Federated Learning (FL) has recently gained attention as an alternative approach for distributed and privacy-aware model training. In FL, devices train machine learning models locally using their own data and share only model updates with the central server instead of transmitting raw data. This decentralized training process helps preserve privacy while also reducing communication costs. Owing to these advantages, federated learning has become an important research direction for building secure and privacy-

preserving intrusion detection systems in distributed IoT environments.

1.5 Limitations of Existing FL IDS

Although federated learning provides significant advantages for privacy-preserving intrusion detection, existing FL-based IDS approaches still face several limitations. Despite its advantages, Federated Learning faces several practical challenges in IoT intrusion detection scenarios. The distributed nature of IoT devices often leads to highly heterogeneous and non-independent identically distributed (non-IID) data across clients. In addition, IoT intrusion datasets frequently contain noisy labels, severe class imbalance, and dynamically changing attack distributions. Most existing Federated Learning-based intrusion detection studies primarily focus on communication efficiency and privacy preservation, while the role of loss-function optimization under heterogeneous federated environments remains comparatively underexplored. Under such conditions, conventional optimization methods based on Binary Cross-Entropy (BCE) loss may produce unstable gradients and degraded convergence performance, ultimately reducing the robustness and generalization capability of the global federated model. Recent studies have shown that traditional sigmoid and cross-entropy based optimization methods often fail to provide sufficient robustness under heterogeneous and noisy learning conditions.

1.6 Motivation for q-Sigmoid Learning

To address these limitations which we mention in the limitation of existing FL IDS, we investigate the integration of q-Sigmoid optimization within Federated Learning for IoT intrusion detection. The q-Sigmoid function, derived from Tsallis non-extensive statistics, introduces a tunable parameter q that controls the shape and sensitivity of the loss function. Unlike conventional sigmoid-based optimization, q-Sigmoid provides improved robustness against noisy samples and heterogeneous data distributions by adaptively modifying gradient behavior during training. This property makes q-Sigmoid particularly suitable for Federated Learning environments where client data distributions vary significantly across devices. Based on Tsallis statistics, the q-exponential and q-logarithmic functions enable the development of generalized activation and loss functions such as q-sigmoid and q-cross-entropy loss. Unlike conventional sigmoid functions, the q-sigmoid formulation allows adaptive control of the decision boundary through the entropic index q , thereby improving robustness against adversarial updates and noisy samples. Motivated by these advantages, this thesis proposes a q-sigmoid enhanced federated learning framework for secure IoT intrusion detection. The proposed method integrates q-sigmoid activation into federated logistic regression to improve detection performance, robustness, and stability under heterogeneous IoT environments. Extensive experimental analysis on

benchmark IoT datasets demonstrates that the proposed framework achieves superior performance compared to traditional federated learning approaches using standard cross-entropy loss functions.

1.7 Problem Statement

Internet of Things (IoT) devices has significantly increased the complexity and vulnerability of modern communication networks [12, 13]. IoT environments are highly susceptible to cybersecurity threats such as denial-of-service attacks, botnet recruitment, malware injection, spoofing attacks, and unauthorized device access [4, 5]. With the rapid growth of IoT networks, maintaining secure communication between connected devices has become increasingly difficult. Large-scale IoT environments often include thousands of devices communicating continuously, making them attractive targets for cyberattacks. Detecting malicious devices and maintaining reliable network security are therefore important requirements for modern IoT systems.

Conventional intrusion detection approaches usually depend on centralized architectures in which traffic data collected from distributed devices is sent to a central server for processing and analysis. Although this approach can simplify monitoring, it also introduces several practical problems. Continuous transmission of large amounts of network traffic increases communication cost, creates latency issues, and raises concerns regarding scalability and user privacy. As the number of connected devices grows, these centralized systems become less efficient for real-time security monitoring in distributed IoT environments.

To overcome some of these limitations, Federated Learning (FL) has recently been explored for IoT security applications. FL allows multiple devices to collaboratively train a shared machine learning model while keeping raw data stored locally on each client device. Since only model parameters or updates are exchanged with the server, federated learning provides a more privacy-aware solution for distributed intrusion detection.

Despite these advantages, practical deployment of FL-based intrusion detection systems remains challenging. IoT devices often generate heterogeneous and non-IID data, where traffic patterns differ significantly from one device to another. In addition, IoT datasets commonly contain noisy samples and severe class imbalance, with attack traffic appearing far less frequently than normal traffic. These factors can negatively affect model convergence and reduce detection accuracy in federated environments.

Another limitation is related to the choice of loss functions used during model training. Most existing federated learning approaches rely on standard loss functions such as cross-entropy or sigmoid loss. While effective in many applications, these loss functions are often sensitive to noisy labels, outliers, and imbalanced data distributions. As a result, model performance may degrade under realistic IoT conditions where data quality and

distribution vary across clients.

Recent research has shown that q-sigmoid loss functions derived from Tsallis entropy can improve robustness against noisy and imbalanced data. By introducing a tunable parameter, the q-sigmoid function provides greater flexibility in controlling the learning behavior of the model and reducing the influence of outliers. However, only a limited number of studies have investigated the use of q-sigmoid-based optimization within federated learning frameworks for IoT intrusion detection.

Motivated by these challenges, this research proposes a privacy-preserving federated learning framework that incorporates q-sigmoid loss for secure IoT network management. The objective is to improve malicious device detection performance while handling noisy, heterogeneous, and imbalanced IoT traffic more effectively.

1.8 Research Motivation

The widespread adoption of Internet of Things (IoT) technology has significantly changed the way devices interact and exchange information in modern digital systems. IoT applications are now present in many domains, including healthcare monitoring, smart homes, industrial control systems, transportation networks, agriculture, and smart city services. These systems improve automation, efficiency, and real-time decision-making by connecting a large number of intelligent devices through the internet. However, the growing dependence on interconnected devices has also introduced major security concerns.

As the number of connected IoT devices increases, networks become more exposed to cyber threats and unauthorized activities. IoT environments are frequently targeted by attacks such as malware injection, denial-of-service attacks, spoofing, botnet-based intrusions, and unauthorized access attempts. Since many IoT devices operate with lightweight hardware and limited security configurations, they are often easier to compromise than traditional computing systems. Protecting IoT networks against these threats has therefore become an important area of research.

Many existing intrusion detection approaches depend on centralized architectures where traffic data from distributed devices is transferred to a central server for analysis. Although centralized learning simplifies model management, it also creates several practical difficulties in large-scale IoT deployments. Constant transmission of traffic data increases communication overhead and network latency, while the collection of sensitive information at a central location introduces privacy risks. In addition, IoT devices continuously generate diverse and high-volume traffic data, making centralized processing computationally demanding and difficult to scale efficiently.

To address these issues, researchers have increasingly explored distributed learning methods that can support privacy-aware model training. Federated Learning (FL) has emerged as one of the most promising approaches in this direction. In FL, devices train

machine learning models locally using their own data and share only model updates with a central server. This decentralized learning process reduces the need to transfer raw data and helps preserve user privacy while lowering communication costs.

Even with these advantages, federated learning in IoT environments still faces several challenges. Data generated by IoT devices is often highly heterogeneous, and different clients may observe completely different traffic patterns. Moreover, intrusion detection datasets usually contain severe class imbalance because malicious traffic samples are much less common than normal traffic. Noisy labels and non-IID data distributions can further affect model convergence and reduce detection performance across distributed clients.

Another challenge comes from the loss functions commonly used in machine learning models. Standard loss functions such as cross-entropy and sigmoid loss may not perform well when the training data contains noise, outliers, or highly imbalanced classes. Under such conditions, the model may become biased toward majority classes and fail to detect malicious activities effectively. Recent studies based on Tsallis entropy have shown that q-sigmoid loss functions can improve robustness against noisy samples and imbalanced datasets by introducing a tunable parameter that controls learning sensitivity. These properties make q-sigmoid-based optimization a promising direction for improving intrusion detection performance in federated IoT environments. However, the integration of q-sigmoid-based optimization within federated learning frameworks for IoT security remains relatively unexplored.

The motivation behind this research is therefore to develop a robust and privacy-preserving federated learning framework capable of improving malicious device detection in complex IoT environments. By integrating the q-sigmoid loss function into the federated learning process, this work aims to enhance robustness against noisy and imbalanced traffic data while preserving user privacy and supporting distributed learning across heterogeneous IoT devices. The proposed approach is expected to contribute toward more secure, scalable, and reliable IoT network management systems..

1.9 Research Gap

Although Federated Learning has gained considerable attention for privacy-preserving intrusion detection, most existing FL-based IDS frameworks continue to rely on conventional optimization functions such as Binary Cross-Entropy, Focal Loss, or Generalized Cross-Entropy. These loss functions often exhibit performance degradation under highly imbalanced and heterogeneous federated environments. Furthermore, limited research has explored the application of q-Sigmoid optimization within Federated Learning for IoT intrusion detection. This research focus on optimizing the parameter q in federated setup. This research shows us two type of method for finding the optimal value of q parameter the first one is using grid search method and the second one is client specific q value de-

pend on dataset imbalance . Recent research based on Tsallis entropy and q-sigmoid loss functions has demonstrated improved robustness against noisy and imbalanced datasets in binary classification problems [14]. However, most of these studies have been evaluated in centralized machine learning environments rather than distributed federated learning systems. Therefore, this research aims to address these gaps by developing a q-sigmoid enhanced federated learning framework for secure IoT network management. The proposed approach focuses on improving robustness against noisy and imbalanced traffic data while preserving privacy and supporting distributed learning across heterogeneous IoT environments.

1.10 Research Objectives

The primary objective of this research is to investigate the effectiveness of the q-sigmoid loss function in federated learning-based IoT intrusion detection systems. The study focuses on analyzing how the tuning of the q parameter influences model performance under distributed and privacy-preserving learning environments.

The specific objectives of this research are as follows:

1. A q-Sigmoid enhanced Federated Learning framework is proposed for privacy-preserving IoT intrusion detection.
2. An adaptive PerClient-q-Sigmoid strategy is introduced to assign different q-values based on client-level attack distributions under heterogeneous federated environments.
3. A validation-based q-sensitivity analysis and threshold optimization strategy is developed for stable q parameter selection without test-data leakage.
4. The proposed framework is evaluated under both IID and non-IID federated settings using benchmark IoT intrusion detection datasets.
5. To compare the performance of q-sigmoid loss with conventional loss functions such as cross-entropy loss and standard sigmoid loss in terms of accuracy, precision, recall, F1-score, and false positive rate.
6. To evaluate the robustness of the proposed approach under noisy and imbalanced IoT traffic datasets.

1.11 Thesis Organisation

This thesis is organised into six chapters, each focusing on different aspects of the proposed q-Sigmoid enhanced Federated Learning framework for secure IoT intrusion detec-

tion.

- **Chapter 1: Introduction** presents the background of IoT security and discusses the growing cybersecurity challenges in distributed IoT environments. The chapter highlights IoT attacks and vulnerabilities, challenges in designing intrusion detection systems, and the role of Machine Learning, Deep Learning, and Federated Learning in IoT security. It further discusses the limitations of existing Federated Learning-based IDS approaches, introduces the motivation for q-Sigmoid learning, defines the research problem, identifies the research gap, and outlines the objectives of the proposed work.
- **Chapter 2: Literature Review** provides a comprehensive review of existing research related to IoT intrusion detection systems, machine learning and deep learning-based IDS frameworks, Federated Learning approaches, adversarial challenges in distributed learning, and entropy-based optimization methods. The chapter also discusses Tsallis entropy, q-Sigmoid optimization, and existing robust loss functions, followed by a comparative analysis of related studies and identification of research gaps addressed in this thesis.
- **Chapter 3: Mathematical Background and Theoretical Foundations** presents the mathematical concepts and theoretical foundations required for the proposed framework. The chapter explains Logistic Regression, Binary Cross-Entropy loss, Generalized Cross-Entropy loss, Focal Loss, Federated Learning optimization, non-IID learning challenges, Tsallis entropy, q-exponential and q-logarithmic functions, and the q-Sigmoid function. The limitations of existing optimization methods are also discussed to motivate the proposed robust learning framework.
- **Chapter 4: q-Sigmoid Enhanced Federated Learning Framework** introduces the proposed Federated Learning framework for secure IoT intrusion detection. The chapter presents the theoretical formulation of Tsallis non-extensive statistics and q-Sigmoid activation, followed by the implementation of Federated Averaging and local training using q-Sigmoid loss. It further introduces the proposed PerClient-q-Sigmoid strategy, adaptive q-value assignment mechanism, learning-rate scaling approach, baseline loss-function comparisons, hyperparameter tuning methodology, and classification threshold optimization.
- **Chapter 5: Results and Discussion** presents the experimental setup, datasets, evaluation metrics, and federated data partitioning strategies used in this research. The chapter analyzes the q-parameter sensitivity, client distribution behavior, IID and non-IID Federated Learning performance, and convergence characteristics of

different optimization methods across benchmark datasets including UNSW-NB15, WUSTL-IIoT-2021, and CSE-CIC-IDS2018. Comparative performance analysis and detailed discussions of the obtained results are also provided.

- **Chapter 6: Conclusion and Future Work** summarizes the major contributions and experimental findings of the thesis. The chapter discusses the effectiveness of the proposed q-Sigmoid enhanced Federated Learning framework in handling noisy, imbalanced, and heterogeneous IoT intrusion data. It also outlines the limitations of the current work and presents several future research directions, including adaptive q-learning, differential privacy integration, deep learning extensions, communication-efficient Federated Learning, and real-world deployment studies.

Chapter 2

Literature Review

2.1 Introduction

Recent advances in Internet of Things (IoT) technologies have significantly increased the demand for intelligent and privacy-preserving cybersecurity frameworks. As IoT devices continuously generate distributed network traffic across heterogeneous environments, intrusion detection systems (IDS) have become essential for identifying malicious activities and protecting critical infrastructure. Traditional centralized intrusion detection approaches often suffer from scalability limitations, communication overhead, and privacy concerns, particularly in large-scale IoT deployments. Consequently, researchers have increasingly explored machine learning, deep learning, and Federated Learning-based intrusion detection frameworks for secure IoT network management. However, the increasing number of interconnected devices has also expanded the cyber-attack surface of modern communication networks. IoT environments are highly vulnerable to Distributed Denial of Service (DDoS) attacks, malware propagation, spoofing attacks, botnet recruitment, ransomware, phishing, and unauthorized access [15, 16]. Traditional security mechanisms are often insufficient for detecting sophisticated and evolving attack patterns in dynamic and heterogeneous IoT environments. Consequently, intrusion detection systems (IDS) based on machine learning, deep learning, and Federated Learning (FL) have emerged as important research areas for intelligent IoT security.

This chapter critically reviews existing research related to IoT intrusion detection systems, machine learning and deep learning-based IDS approaches, Federated Learning-based intrusion detection frameworks, adversarial challenges in distributed learning systems, and entropy-based optimization techniques including Tsallis statistics and q-sigmoid learning. The chapter further analyzes the limitations of existing methods and identifies the research gap addressed by the proposed q-sigmoid enhanced Federated Learning framework.

2.2 IoT Intrusion Detection Systems

Intrusion Detection Systems (IDS) are security frameworks designed to monitor network traffic and identify malicious activities or unauthorized access attempts within communication systems. IDS techniques are generally categorized into signature-based IDS and anomaly-based IDS.

Signature-based IDS approaches identify attacks by comparing network traffic patterns with predefined attack signatures. These methods achieve high detection accuracy for known attacks but fail to detect zero-day attacks and evolving threat patterns. In contrast, anomaly-based IDS systems identify deviations from normal network behavior using statistical analysis and machine learning techniques. Although anomaly-based methods can detect unknown attacks, they often suffer from high false positive rates.

Aziz et al. [15] reviewed emerging cyber-security threats in IoT environments including DDoS attacks, malware propagation, spoofing attacks, botnet recruitment, and unauthorized device access. Their study highlighted the increasing complexity of IoT attack surfaces and emphasized the need for intelligent intrusion detection frameworks capable of handling dynamic network behavior.

Lashkari et al. [16] analyzed machine learning-based intrusion detection approaches for IoT systems and demonstrated that intelligent traffic analysis significantly improves attack detection performance compared to traditional rule-based security systems.

Despite these advances, conventional IDS frameworks still face several limitations in IoT environments due to:

- Resource-constrained IoT devices
- Heterogeneous communication protocols
- Dynamic traffic behavior
- Scalability limitations
- Privacy concerns

Although conventional IDS techniques have demonstrated effectiveness in detecting known attack patterns, their performance often degrades in dynamic IoT environments characterized by heterogeneous traffic distributions, evolving attack behaviors, and resource-constrained edge devices. Furthermore, most traditional IDS frameworks rely on centralized architectures that introduce privacy risks and communication overhead during large-scale deployment.

2.3 Machine Learning-Based Intrusion Detection Systems

Machine learning techniques have become widely adopted for intrusion detection because they can automatically learn traffic behavior patterns and classify malicious activities with improved accuracy.

Recent studies have demonstrated that machine learning-based IDS frameworks significantly improve intrusion detection performance compared to traditional rule-based systems. Algorithms such as Random Forest, Support Vector Machine (SVM), Logistic Regression, and Artificial Neural Networks (ANN) have shown strong classification capability on benchmark datasets including UNSW-NB15 and CICIDS2017. However, many machine learning-based IDS systems remain dependent on centralized training architectures, making them vulnerable to privacy leakage, scalability limitations, and communication bottlenecks in distributed IoT environments.

The centralized collection of traffic data introduces:

- Communication overhead
- Privacy leakage
- Scalability limitations
- Increased latency

These limitations motivated the adoption of Federated Learning for privacy-preserving distributed intrusion detection.

2.4 Deep Learning-Based IDS

Deep learning has emerged as a powerful approach for IoT intrusion detection because of its ability to automatically extract complex traffic features from large-scale network datasets.

CNN-based intrusion detection systems are effective for identifying spatial traffic patterns and have demonstrated high detection accuracy on datasets such as UNSW-NB15 and Bot-IoT. Similarly, LSTM and RNN models are effective for sequential traffic analysis and temporal attack detection.

Several researchers have proposed hybrid CNN-LSTM architectures for improving feature extraction and attack classification performance. These approaches generally outperform traditional machine learning methods in terms of accuracy, recall, and F1-score.

Although deep learning significantly improves intrusion detection capability, these approaches require:

- Large-scale labeled datasets
- High computational resources
- Expensive centralized training

Despite achieving high detection accuracy, deep learning-based IDS approaches generally require large-scale centralized datasets and computationally intensive training procedures. These limitations reduce their suitability for resource-constrained IoT environments. Additionally, most deep learning IDS frameworks do not adequately address privacy preservation or heterogeneous distributed learning scenarios.

2.5 Federated Learning-Based Intrusion Detection Systems

Federated Learning (FL) is a distributed machine learning paradigm in which multiple clients collaboratively train a shared global model without exchanging raw data. Instead, each client performs local training using private datasets and transmits only model parameters or gradients to a centralized aggregation server.

McMahan et al. [17] introduced the Federated Averaging (FedAvg) algorithm, which became one of the foundational optimization techniques in Federated Learning systems.

Federated Learning has become highly suitable for IoT environments because it improves privacy preservation while reducing communication overhead.

Although Federated Learning improves privacy preservation by avoiding direct data sharing, existing FL-based intrusion detection systems continue to face several unresolved challenges. These include statistical heterogeneity across clients, convergence instability under non-IID data distributions, noisy local labels, class imbalance, and vulnerability to adversarial client updates. Furthermore, most existing FL-based IDS frameworks rely on conventional Binary Cross-Entropy optimization without investigating the influence of alternative robust loss functions under heterogeneous federated environments.

2.6 Adversarial Challenges in Federated Learning

Despite its privacy-preserving advantages, Federated Learning introduces several new security vulnerabilities in distributed learning systems.

Bagdasaryan et al. [18] demonstrated that malicious clients can inject backdoor attacks into Federated Learning systems through manipulated local model updates. Their work highlighted the vulnerability of FedAvg aggregation under adversarial conditions.

Pillutla et al. [19] proposed robust aggregation methods for Federated Learning under Byzantine attack settings. Their study showed that malicious gradients can significantly degrade global model performance if robust aggregation mechanisms are not employed.

Adversarial attacks in Federated Learning systems generally include:

- Model poisoning attacks
- Data poisoning attacks
- Byzantine attacks
- Backdoor attacks
- Gradient manipulation attacks

These attacks are particularly harmful in IoT environments because compromised devices may continuously transmit malicious model updates during collaborative training. These adversarial and heterogeneous learning challenges highlight the importance of robust optimization strategies capable of stabilizing federated training under noisy and imbalanced client environments. Consequently, recent research has explored robust entropy-based optimization methods to improve learning stability and resilience in distributed machine learning systems.

2.7 Entropy-Based Learning and Tsallis Statistics

Entropy-based optimization methods derived from Tsallis non-extensive statistics have attracted increasing attention because of their robustness against noisy samples, outliers, and heavy-tailed data distributions. Unlike conventional entropy formulations, Tsallis [20] introduced Tsallis entropy as a generalized extension of Shannon entropy based on non-extensive statistical mechanics. Tsallis entropy incorporates a tunable entropic parameter q that controls the degree of non-extensivity and robustness against heavy-tailed distributions. Tsallis entropy introduces a tunable entropic parameter q that adaptively controls optimization sensitivity. These properties are particularly relevant for IoT intrusion detection, where network traffic distributions are often highly imbalanced and heterogeneous. Entropy-based optimization methods have attracted significant attention for improving robustness in noisy and uncertain learning environments.

The Tsallis entropy formulation is defined as:

$$H_q(p) = \frac{1 - \sum_i p_i^q}{q - 1}$$

where q is the entropic index.

Tsallis entropy has been widely explored for robust optimization under noisy and imbalanced data distributions because it provides adaptive sensitivity to outliers and minority samples.

2.8 q-Sigmoid Optimization and Robust Loss Functions

Recent studies on q-sigmoid optimization have demonstrated promising robustness properties in noisy and imbalanced binary classification tasks. The q-sigmoid function extends the conventional sigmoid activation using Tsallis q-exponential functions, enabling adaptive gradient behavior during optimization. Compared with standard Binary Cross-Entropy loss, q-sigmoid optimization can reduce sensitivity to outliers and improve convergence stability under heterogeneous data distributions.

To improve optimization robustness, researchers have explored generalized entropy-based loss functions derived from Tsallis statistics.

Pardede et al. [21] proposed a generalized q-sigmoid loss function for binary classification tasks. Their study demonstrated that q-sigmoid optimization improves robustness against noisy labels and outliers compared to conventional sigmoid loss functions.

The q-sigmoid activation is defined as:

$$\sigma_q(x) = \frac{1}{1 + \exp_q(-x)}$$

where the q-exponential function is given as:

$$\exp_q(x) = [1 + (1 - q)x]^{\frac{1}{1-q}}$$

The deformation parameter q controls:

- Optimization sensitivity
- Noise robustness
- Outlier resistance
- Classification stability

Experimental results presented by Pardede et al. demonstrated improved classification performance under noisy learning conditions. However, their work primarily focused on centralized machine learning environments rather than Federated Learning systems.

However, most existing q-sigmoid studies have been evaluated primarily in centralized machine learning environments. Very limited research has investigated the integration of q-sigmoid optimization within Federated Learning architectures for secure IoT intrusion detection under non-IID client distributions. This limitation motivates the development of the proposed q-sigmoid enhanced Federated Learning framework presented in this thesis

2.9 Comparative Analysis of Existing Studies

Table 2.1 Comparative analysis of existing IoT intrusion detection and Federated Learning approaches with corresponding limitations and research gaps.

Table 2.1: Comparative Analysis of Existing IoT Intrusion Detection and Federated Learning Approaches

Reference	Approach	Dataset	Learning Type	Loss Function	Limitations
Rahman et al. [22]	Federated IDS for IoT	IoT Traffic	Federated Learning	Cross-Entropy	Performance degradation under non-IID client distributions and communication overhead issues.
Zhang et al. [23]	FL-based IoT Security	Multiple IoT Datasets	Federated Learning	Cross-Entropy	Did not address noisy labels and adversarial robustness in heterogeneous environments.
Villanueva et al. [24]	Real-time IoT Anomaly Detection	IoT Dataset	Federated Learning	BCE Loss	Sensitive to noisy traffic samples and unstable under imbalanced datasets.
Khraisat et al. [25]	Privacy-Preserving FL IDS	IoT Dataset	Federated Learning	Sigmoid Loss	Sensitive to heterogeneous client distributions.
Bagdasaryan et al. [18]	Backdoor Attacks in FL	Federated Systems	Federated Learning	Standard Optimization	Vulnerable to malicious client updates and poisoning attacks.
Pillutla et al. [19]	Robust Aggregation for FL	Distributed Data	Federated Learning	Standard Loss	Increased computational complexity and aggregation overhead.
Lashkari et al. [16]	ML-Based IoT IDS	UNSW-NB15	Centralized Learning	Cross-Entropy	Centralized architecture introduces privacy and scalability limitations.
Gugueoth et al. [26]	Intelligent IoT IDS	IoT Security Data	Centralized Learning	Standard Sigmoid	Did not address distributed learning or noisy label robustness.

Reference	Approach	Dataset	Learning Type	Loss Function	Limitations
Pardede et al. [21]	q-Sigmoid Optimization	Binary Data	Centralized Learning	q-Sigmoid Loss	Experiments limited to centralized learning environments.
Tsallis [20]	Tsallis Entropy	Theoretical Study	Statistical Learning	Entropy-Based	Did not investigate IDS or Federated Learning applications.
Proposed Work	q-Sigmoid Enhanced FL IDS	UNSW-NB15, WUSTL-IIoT-2021,IDS2018	Non-IID Federated Learning	q-Sigmoid Loss	Improves robustness against noisy labels, class imbalance, and heterogeneous IoT traffic while preserving privacy.

Table 2.1 summarizes the limitations of existing intrusion detection and Federated Learning approaches for IoT security. The comparative analysis indicates that although Federated Learning improves privacy preservation, existing methods still suffer from optimization instability, noisy label sensitivity, non-IID data challenges, and class imbalance issues. Furthermore, very limited research has investigated the integration of q-sigmoid optimization within Federated Learning environments for secure IoT intrusion detection. These limitations motivate the proposed q-sigmoid enhanced Federated Learning framework presented in this thesis.

2.10 Research Gap Analysis

The literature review indicates that significant progress has been made in IoT intrusion detection using machine learning, deep learning, and Federated Learning approaches. However, several important limitations remain unresolved.

- Non-IID client distributions
- Noisy labels
- Class imbalance
- Device heterogeneity
- Optimization instability
- Adversarial client attacks

Furthermore, most Federated Learning-based IDS frameworks utilize conventional Binary Cross-Entropy and standard sigmoid loss functions, which are highly sensitive to noisy and imbalanced IoT traffic datasets.

Recent studies on q-sigmoid optimization have demonstrated improved robustness against noisy labels and outliers; however, these methods have primarily been evaluated in centralized machine learning environments.

To the best of our knowledge, very limited research has investigated the integration of Tsallis entropy-based q-sigmoid optimization within Federated Learning frameworks for secure IoT intrusion detection under heterogeneous and non-IID traffic environments. Therefore, this thesis proposes a q-sigmoid enhanced Federated Learning framework designed to improve robustness against noisy labels, class imbalance, and heterogeneous client distributions while preserving privacy in distributed IoT intrusion detection environments.

2.11 Chapter Summary

This chapter critically reviewed existing research related to IoT intrusion detection systems, machine learning and deep learning-based IDS approaches, Federated Learning frameworks, adversarial challenges in distributed learning systems, and entropy-based optimization techniques derived from Tsallis statistics. The review identified several limitations in existing approaches, including sensitivity to noisy labels, instability under non-IID client distributions, privacy concerns, and limited investigation of robust loss-function optimization within Federated Learning environments. These observations motivate the proposed q-sigmoid enhanced Federated Learning framework presented in the subsequent chapters.

Chapter 3

Mathematical Background and Theoretical Foundations

3.1 Introduction

This chapter presents the mathematical foundations and theoretical concepts required for the proposed q-sigmoid enhanced Federated Learning framework for IoT intrusion detection. The chapter discusses Logistic Regression, Binary Cross-Entropy (BCE) loss, Generalized Cross-Entropy (GCE) loss, Focal Loss, Federated Learning optimization, non-IID learning challenges, Tsallis entropy, q-exponential functions, q-logarithmic functions, and q-sigmoid optimization. These theoretical concepts establish the mathematical basis for the proposed robust intrusion detection framework presented in the subsequent chapters.

Machine learning and Federated Learning optimization rely heavily on mathematical objective functions that guide model convergence during training. The performance of intrusion detection systems depends not only on feature extraction and classification capability but also on the robustness and stability of optimization procedures under noisy, imbalanced, and heterogeneous data distributions.

In practical IoT environments, intrusion detection datasets frequently contain noisy labels, minority attack samples, and non-identically distributed client traffic patterns. Consequently, conventional optimization methods often experience degraded convergence and unstable learning behavior. To address these challenges, researchers have explored robust optimization techniques derived from non-extensive statistical mechanics. Among these approaches, Tsallis entropy and q-sigmoid optimization have attracted significant attention because of their adaptive robustness properties under noisy and heterogeneous learning environments.

3.2 Logistic Regression

Logistic Regression is one of the most widely used supervised learning algorithms for binary classification problems. It estimates the probability that a given input belongs

to a particular class using the sigmoid activation function. Because intrusion detection systems generally classify network traffic into benign or malicious categories, Logistic Regression is highly suitable for binary attack detection.

For an input feature vector $x \in \mathbb{R}^d$, the linear model is represented as:

$$z = w^T x + b$$

where:

- w represents the weight vector,
- b denotes the bias term,
- x is the input feature vector.

The sigmoid activation function transforms the linear output into a probability value:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The predicted probability is expressed as:

$$P(y = 1|x) = \sigma(w^T x + b)$$

Logistic Regression attempts to identify an optimal decision boundary separating malicious and benign traffic samples. During optimization, model parameters are updated iteratively to minimize classification loss functions [27].

3.3 Binary Cross-Entropy Loss Function

Binary Cross-Entropy (BCE) loss is one of the most commonly used objective functions for binary classification tasks. BCE measures the difference between predicted probabilities and actual class labels.

The BCE loss function is mathematically defined as:

$$L_{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

where:

- N denotes the number of training samples,
- y_i represents the true label,
- $\hat{y}_i$ denotes the predicted probability.

Although BCE performs effectively under balanced and clean datasets, it exhibits several limitations in intrusion detection systems including sensitivity to noisy labels and poor minority class learning capability [28].

3.4 Generalized Cross-Entropy Loss

Generalized Cross-Entropy (GCE) loss was introduced to improve robustness against noisy labels. GCE combines the advantages of Mean Absolute Error and Cross-Entropy loss.

The GCE loss function is represented as:

$$L_{GCE} = \frac{1 - \hat{y}_i^q}{q}$$

where:

- $q \in (0, 1]$ controls the robustness behavior of the loss function.

When $q \rightarrow 0$, the GCE formulation approaches conventional Cross-Entropy loss. GCE reduces the influence of noisy samples during optimization and improves robustness under corrupted datasets [29].

3.5 Focal Loss Function

Focal Loss was proposed to address severe class imbalance problems in classification tasks. Unlike BCE loss, Focal Loss reduces the contribution of easily classified samples and focuses training on difficult examples.

The Focal Loss function is mathematically represented as:

$$\mathcal{L}_{\text{Focal}}(y, p) = -[\alpha y(1 - p)^\gamma \log p + (1 - \alpha)(1 - y)p^\gamma \log(1 - p)] \quad (3.1)$$

We use $\gamma = 2.0$ and $\alpha = 0.25$ following Lin et al. [30]

where:

- γ is the focusing parameter.
- α is the focusing parameter.
- p is the predicted probability for true class.

3.6 Federated Learning

Federated Learning (FL) is a distributed machine learning paradigm in which multiple clients collaboratively train a shared global model without exchanging raw data.

Suppose there are K participating clients. The global optimization objective can be expressed as:

$$F(w) = \sum_{k=1}^K \frac{n_k}{n} F_k(w)$$

where:

- $F_k(w)$ denotes the local objective function of client k ,
- n_k represents the number of samples at client k ,
- $n = \sum_{k=1}^K n_k$ denotes the total number of samples.

The Federated Averaging (FedAvg) algorithm aggregates local model parameters using weighted averaging:

$$w_{t+1} = \sum_{k=1}^K \frac{n_k}{n} w_k^{(t)}$$

Federated Learning improves privacy preservation because raw traffic data remains localized at client devices [17].

3.7 Non-IID Learning in Federated Environments

In practical IoT environments, client data distributions are highly heterogeneous and non-identically distributed (non-IID). Different IoT devices generate diverse traffic patterns depending on communication protocols, device type, and user behavior.

Under non-IID conditions, client distributions differ significantly:

$$P_k(x, y) \neq P_j(x, y)$$

where:

- $P_k(x, y)$ denotes the data distribution of client k ,
- $P_j(x, y)$ denotes the data distribution of client j .

Non-IID learning introduces several optimization challenges including:

- Client drift
- Slow convergence
- Model instability
- Reduced generalization performance

These challenges significantly affect Federated Learning convergence and intrusion detection performance [31].

3.8 Entropy-based Intrusion Detection

In parallel ,entropy-based methods have been extensively explored for intrusion detection due to their strong capacity to quantify uncertainty and capture variations in data distributions. These methods rely on measuring the randomness or disorder in network traffic,enabling the detection of anomalous patterns without requiring prior knowledge of attack signatures. this makes them particularly suitable for identifying previously unseen or evolving attacks in dynamic network environments. Early works in network security demonstrated the effectiveness of entropy- based analysis for anamoly detection

3.9 Tsallis Entropy

Tsallis entropy was introduced as a generalized extension of Shannon entropy based on non-extensive statistical mechanics.

The Tsallis entropy formulation is expressed as:

$$H_q(p) = \frac{1 - \sum_i p_i^q}{q - 1}$$

where:

- p_i denotes the probability distribution,
- q represents the entropic index parameter.

When $q \rightarrow 1$, Tsallis entropy reduces to classical Shannon entropy:

$$H(p) = - \sum_i p_i \log p_i$$

Tsallis entropy provides adaptive robustness against noisy labels, outliers, and heterogeneous learning environments [20].

3.10 q-Exponential and q-Logarithmic Functions

The q-exponential function derived from Tsallis statistics is defined as:

$$\exp_q(x) = [1 + (1 - q)x]_+^{\frac{1}{1-q}}$$

Similarly, the q-logarithmic function is represented as:

$$\ln_q(x) = \frac{x^{1-q} - 1}{1 - q}$$

These generalized mathematical functions provide adaptive optimization behavior and flexible sensitivity control under noisy and heterogeneous learning environments.

3.11 q-Sigmoid Function

The q-sigmoid function is a generalized sigmoid activation function derived from Tsallis statistics. It replaces the standard exponential function with the q-exponential formulation.

The q-sigmoid function is mathematically defined as:

$$\sigma_q(x) = \frac{1}{1 + \exp_q(-x)}$$

where:

- q is the deformation parameter controlling optimization sensitivity.

Compared to the standard sigmoid function, q-sigmoid provides:

- Improved robustness against noisy labels
- Better handling of class imbalance
- Adaptive optimization sensitivity
- Reduced influence of outliers

These properties make q-sigmoid suitable for intrusion detection systems involving noisy IoT traffic and heterogeneous Federated Learning environments [21].

3.12 Gradient Optimization in Logistic Regression

Gradient-based optimization methods are widely used for minimizing classification loss functions in machine learning systems.

For Logistic Regression using BCE loss, the gradient with respect to model parameters is represented as:

$$\frac{\partial L}{\partial w} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i) x_i$$

Gradient Descent updates model parameters iteratively using:

$$w_{t+1} = w_t - \eta \frac{\partial L}{\partial w}$$

where:

- η is the learning rate.

Under noisy and heterogeneous Federated Learning environments, gradients may become unstable because of noisy labels, outliers, and non-IID traffic distributions.

3.13 Limitations of Existing Optimization Methods

Although Binary Cross-Entropy, Generalized Cross-Entropy, and Focal Loss functions have demonstrated strong performance in conventional classification problems, these methods still suffer from several limitations in Federated Learning environments involving heterogeneous IoT traffic distributions.

Binary Cross-Entropy loss is highly sensitive to noisy labels and minority attack samples. Similarly, Focal Loss primarily addresses class imbalance but may remain unstable under non-IID Federated Learning settings. Generalized Cross-Entropy improves robustness against noisy labels; however, its optimization behavior under distributed IoT environments remains limited.

Furthermore, existing Federated Learning optimization methods often experience convergence instability under heterogeneous client distributions and adversarial traffic conditions. These limitations motivate the development of a robust q-sigmoid enhanced Federated Learning framework capable of improving intrusion detection performance under noisy and imbalanced IoT environments.

3.14 Chapter Summary

This chapter presented the mathematical foundations required for the proposed q -sigmoid enhanced Federated Learning framework for IoT intrusion detection. The chapter discussed Logistic Regression, Binary Cross-Entropy loss, Generalized Cross-Entropy loss, Focal Loss, Federated Learning optimization, non-IID learning challenges, Tsallis entropy, q -exponential functions, q -logarithmic functions, and q -sigmoid optimization.

The theoretical concepts presented in this chapter establish the mathematical basis for the proposed robust intrusion detection framework described in the next chapter.

Chapter 4

q-Sigmoid Enhanced Federated Learning Framework

4.1 Introduction to the Proposed Framework

This chapter presents the q-sigmoid enhanced Federated Learning framework for robust IoT intrusion detection under heterogeneous and non-IID environments. The proposed framework integrates Tsallis entropy-based q-sigmoid optimization into Federated Learning to improve robustness against noisy labels, class imbalance, and unstable client distributions. Conventional Federated Learning approaches for intrusion detection generally utilize Binary Cross-Entropy (BCE) or standard sigmoid loss functions. However, these optimization methods often suffer from convergence instability under noisy and heterogeneous IoT traffic environments. To address these limitations, this research proposes a robust q-sigmoid optimization framework capable of improving distributed intrusion detection performance while preserving client privacy. Federated Learning (FL) addresses this by enabling edge nodes to train local models and share only model parameters, never raw data, with a central aggregation server.

However, two fundamental challenges remain unsolved in FL-based intrusion detection systems. First, the loss functions used during local training govern convergence quality and sensitivity to class imbalance, yet this choice has received limited attention in the FL context. Second, the *statistical heterogeneity* of IoT edge nodes commonly referred to as the non-IID (non-Independent and Identically Distributed) problem causes client gradients to diverge, degrading global model quality.

This chapter presents a comprehensive framework that addresses both challenges through two contributions:

1. **q-Sigmoid Loss for Federated Learning.**

We replace the standard Binary Cross-Entropy (BCE) loss with the Tsallis statistics-based q-Sigmoid activation and q-Cross Entropy loss function. The entropic index q generalises BCE (recovered exactly at $q = 1$) and modulates the gradient tail behaviour via a power-law substitution of the exponential function. The optimal q is selected through a rigorous grid search on a held-out validation

set, ensuring no information leakage from the test set.

2. **PerClient-q-Sigmoid** We propose a mechanism in which each federated client independently computes its own entropic index q_k from its local label distribution prior to training. Clients whose local data is severely skewed towards attacks (or towards normal traffic) receive a higher q_k , amplifying the gradient signal for hard minority examples. The aggregation server performs standard Federated Averaging (FedAvg) [32] without modification, making this contribution plug-and-play on any existing FL system.

The framework is evaluated against four baseline loss functions BCE (Binary Cross-Entropy), Generalised Cross-Entropy (GCE) [33], Focal Loss [30], and grid-tuned q-Sigmoid across both IID and non-IID partitions on three publicly available IoT security datasets: UNSW-NB15 [34] and WUSTL-IIoT-2021 [35] and CSE-CIC-IDS2018 [36]

4.2 Theoretical Background

4.2.1 Tsallis Non-Extensive Statistics

Tsallis statistics [37] generalise the Boltzmann-Gibbs entropy framework by introducing a single real-valued parameter $q \in \mathbb{R}$ (the *entropic index*) that interpolates between sub-extensive ($q < 1$), extensive ($q = 1$), and super-extensive ($q > 1$) regimes. Two core functions underpin the framework:

q-Logarithm

The q -logarithm is defined as:

$$\ln_q(x) = \frac{x^{1-q} - 1}{1 - q}, \quad x > 0, \quad q \neq 1 \quad (4.1)$$

At the limit $q \rightarrow 1$, l'Hôpital's rule recovers the natural logarithm: $\ln_1(x) = \ln(x)$.

q-Exponential

The inverse function, the q -exponential, is:

$$\exp_q(x) = [1 + (1 - q)x]_+^{\frac{1}{1-q}} \quad (4.2)$$

where $[\cdot]_+ = \max(\cdot, 0)$ ensures the base remains non-negative. At $q \rightarrow 1$, $\exp_q(x) \rightarrow e^x$.

The q -exponential replaces the standard exponential in the sigmoid activation, yielding a power-law tail rather than an exponential tail. This is particularly relevant for

IoT intrusion data, where attack traffic distributions are heavy-tailed and deviate from sub-Gaussian assumptions.

4.2.2 q-Sigmoid Activation Function

Substituting the q -exponential into the standard sigmoid definition produces the q -Sigmoid activation:

$$\sigma_q(z) = \frac{1}{1 + \exp_q(-z)} \quad (4.3)$$

Its derivative with respect to the pre-activation z , evaluated at $p = \sigma_q(z)$, takes the form:

$$\frac{d\sigma_q}{dz} = p^q(1 - p)^q \quad (4.4)$$

which reduces to the standard $p(1 - p)$ at $q = 1$. For $q > 1$, the derivative is suppressed near $p = 0$ and $p = 1$ relative to $q = 1$, providing a softer gradient in high-confidence regions and larger relative gradient for uncertain predictions.

4.2.3 q-sigmoid loss

The q -sigmoid loss function is derived by replacing the natural logarithm in BCE with the q -logarithm:

$$\mathcal{L}_q(y, p) = -[y \cdot \ln_q(p) + (1 - y) \cdot \ln_q(1 - p)] \quad (4.5)$$

with L2 regularisation term $\frac{\lambda}{2}\|\mathbf{w}\|^2$ added to the mean loss over a mini-batch. In this work, we also apply class-weight correction:

$$\mathcal{L}_q^{\text{weighted}}(y, p) = -w_y[y \cdot \ln_q(p) + (1 - y) \cdot \ln_q(1 - p)] \quad (4.6)$$

where $w_y = \omega$ for the positive (attack) class and $w_y = 1$ for the negative (normal) class, with $\omega = n_{\text{normal}}/n_{\text{attack}}$ computed from the local training shard. This directly corrects for class imbalance in the gradient signal.

Gradient Derivation

The gradient of $\mathcal{L}_q$ with respect to the pre-activation z decomposes via the chain rule as:

$$\frac{\partial \mathcal{L}_q}{\partial z} = \frac{\partial \mathcal{L}_q}{\partial p} \cdot \frac{\partial p}{\partial z} \quad (4.7)$$

The partial derivative of the loss with respect to p is:

$$\frac{\partial \mathcal{L}_q}{\partial p} = -[y \cdot p^{-q} - (1 - y) \cdot (1 - p)^{-q}] \quad (4.8)$$

Combining with the q-sigmoid derivative (Equation 4.4), the per-sample error signal is:

$$\delta_i = w_{y_i} \cdot [- (y_i p_i^{-q} - (1 - y_i)(1 - p_i)^{-q})] \cdot p_i^q (1 - p_i)^q \quad (4.9)$$

The weight update equations are:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \frac{1}{n} \mathbf{X}^\top \boldsymbol{\delta} + \lambda \mathbf{w} \quad (4.10)$$

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{1}{n} \sum_{i=1}^n \delta_i \quad (4.11)$$

4.3 Federated Learning Framework

The proposed framework follows a standard parameter-server Federated Learning (FL) topology comprising K distributed federated clients (IoT edge devices or gateways) and a central aggregation server. The framework operates in multiple communication rounds where each client performs local model training using its private intrusion detection data. Instead of transmitting raw data, only locally updated model parameters are shared with the central server, thereby preserving data privacy and reducing communication risks.

4.3.1 Federated Averaging (FedAvg)

Following McMahan et al. [32], the server aggregates client parameters using weighted averaging:

$$\mathbf{w}_{\text{global}}^{(t+1)} = \sum_{k=1}^K \frac{n_k}{n_{\text{total}}} \mathbf{w}_k^{(t)} \quad (4.12)$$

where n_k is the number of training samples at client k and $n_{\text{total}} = \sum_k n_k$. The bias is aggregated analogously. This weighting corrects for imbalanced dataset sizes across clients.

4.3.2 Local Training with q-Sigmoid Loss

In each communication round t :

1. The server broadcasts the current global parameters $\mathbf{w}_{\text{global}}^{(t)}$ to all clients.

2. Each client k initialises its local model with $\mathbf{w}_{\text{global}}^{(t)}$ and performs E local gradient descent steps on its private dataset $\mathcal{D}_k$:

$$\mathbf{w}_k \leftarrow \mathbf{w}_k - \eta \nabla_{\mathbf{w}} \mathcal{L}_q^{\text{weighted}}(\mathbf{w}_k; \mathcal{D}_k) \quad (4.13)$$

3. The client returns updated parameters $\mathbf{w}_k^{(t)}$ to the server.
4. The server performs FedAvg aggregation (Equation 4.12).
5. The server tunes the classification threshold τ^* on a held-out validation set $\mathcal{D}_{\text{val}}$:

$$\tau^* = \arg \max_{\tau \in [0.05, 0.95]} \text{F1}(\mathbf{w}_{\text{global}}^{(t+1)}, \mathcal{D}_{\text{val}}, \tau) \quad (4.14)$$

The complete algorithm is presented in Algorithm 1.

Algorithm 1 Federated Learning with q-Sigmoid Loss and PerClient-q Assignment

Require: Clients $\{(k, \mathcal{D}_k)\}_{k=1}^K$, validation set $\mathcal{D}_{\text{val}}$, test set $\mathcal{D}_{\text{test}}$

Require: Grid $\mathcal{Q}$, rounds T , local epochs E , learning rate η , regularisation λ , class weight ω

Ensure: Trained global model $\mathbf{w}^*$, tuned threshold τ^*

Phase 1: q Selection via Offline Grid Search

- 1: **for** each candidate $q \in \mathcal{Q}$ **do**
- 2: Initialise $\mathbf{w} \leftarrow \mathbf{0}$
- 3: **for** $t = 1$ to T **do**
- 4: **for** each client k **do**
- 5: $\mathbf{w}_k \leftarrow \mathbf{w} - \eta \nabla \mathcal{L}_q^{\omega}(\mathbf{w}; \mathcal{D}_k)$ $\triangleright E$ local steps
- 6: **end for**
- 7: $\mathbf{w} \leftarrow \sum_k \frac{n_k}{n_{\text{total}}} \mathbf{w}_k$ $\triangleright$ FedAvg
- 8: **end for**
- 9: $\tau_q \leftarrow \arg \max_{\tau} \text{F1}(\mathbf{w}, \mathcal{D}_{\text{val}}, \tau)$
- 10: $s_q \leftarrow \text{F1}(\mathbf{w}, \mathcal{D}_{\text{val}}, \tau_q)$
- 11: **end for**
- 12: $q^* \leftarrow \arg \max_{q \in \mathcal{Q}} s_q$ $\triangleright$ Best q on validation

Phase 2: Final Training with Best q

- 13: Initialise $\mathbf{w}^* \leftarrow \mathbf{0}$
 - 14: **for** $t = 1$ to T **do**
 - 15: **for** each client k **do**
 - 16: $\mathbf{w}_k \leftarrow \mathbf{w}^* - \eta \nabla \mathcal{L}_{q^*}^{\omega}(\mathbf{w}^*; \mathcal{D}_k)$
 - 17: **end for**
 - 18: $\mathbf{w}^* \leftarrow \sum_k \frac{n_k}{n_{\text{total}}} \mathbf{w}_k$
 - 19: $\tau^* \leftarrow \arg \max_{\tau} \text{F1}(\mathbf{w}^*, \mathcal{D}_{\text{val}}, \tau)$
 - 20: **end for**
 - 21: **return** $\mathbf{w}^*, \tau^*$
-

4.4 Proposed Method: PerClient-q-Sigmoid

4.4.1 Motivation

In non-IID FL, each client k holds a dataset $\mathcal{D}_k$ whose class distribution may differ substantially from the global distribution. A client monitoring a botnet-infected subnet might observe 97% attack traffic, while a gateway in a corporate office may see fewer than 25% attacks. Applying the same loss function $\mathcal{L}_q$ with a globally uniform q to all clients ignores this heterogeneity.

We argue that *the optimal q for a given client is a function of that client’s local class imbalance*: more skewed distributions require a heavier-tailed loss (higher q) to maintain gradient signal for the minority class, while balanced distributions are best served by $q \approx 1$ (approaching BCE). This motivates a per-client assignment of the entropic index.

4.4.2 Per-Client q Assignment Mechanism

Unlike globally tuned q-Sigmoid, the proposed PerClient-q-Sigmoid framework assigns client-specific q -values according to local attack distributions. Each client performs local optimization using its own q parameter, while the server aggregates only model parameters using FedAvg. Therefore, no single global q -value is enforced during federated optimization.

For client k , define the local attack rate $\rho_k = \frac{1}{n_k} \sum_{i \in \mathcal{D}_k} y_i$. The local imbalance is:

$$\psi_k = 2 |\rho_k - 0.5| \in [0, 1] \quad (4.15)$$

where $\psi_k = 0$ denotes a perfectly balanced client and $\psi_k = 1$ denotes a fully homogeneous client. The per-client entropic index is:

$$q_k = q_{\min} + (q_{\max} - q_{\min}) \cdot \sqrt{\psi_k} \quad (4.16)$$

The square-root compression $\sqrt{\psi_k}$ is deliberate: it prevents extreme clients (e.g. $\rho_k = 0.97$, $\psi_k \approx 0.94$) from receiving very large q_k values that would cause their gradients to dominate and conflict with balanced clients during FedAvg aggregation. In our implementation, $q_{\min} = 1.0$ and $q_{\max} = 1.8$.

Table 4.1 illustrates the mapping for representative attack rates.

4.4.3 Learning Rate Scaling

A further complication in heterogeneous- q FL is that clients with higher q produce gradient updates with larger magnitude (due to the steeper δ_i in Equation 4.9). If all clients use the same learning rate η , high- q clients take disproportionately large weight steps

Table 4.1: Per-client q assignment examples ($q_{\min} = 1.0$, $q_{\max} = 1.8$)

Attack Rate ρ_k	Imbalance ψ_k	$\sqrt{\psi_k}$	q_k	Interpretation
50%	0.00	0.00	1.00	Balanced $\rightarrow$ equivalent to BCE
68%	0.36	0.60	1.48	Moderate imbalance (UNSW-NB15 IID)
82%	0.64	0.80	1.64	High imbalance
97%	0.94	0.97	1.78	Extreme imbalance
25%	0.50	0.71	1.57	Minority attack client

and bias the FedAvg result. To compensate, each client’s effective learning rate is scaled inversely with q_k :

$$\eta_k = \frac{\eta}{q_k} \quad (4.17)$$

This ensures that the magnitude of parameter updates is approximately equalised across clients with different q_k values, stabilising the aggregation.

4.4.4 Zero Server-Side Overhead

A key property of PerClient- q -Sigmoid is that it requires *no changes to the server*. The server executes standard FedAvg at every round, agnostic to the q_k values used by each client. The heterogeneous loss functions affect only the local gradient directions; once parameters are returned to the server, aggregation proceeds identically to any standard FL round. This makes the method a drop-in enhancement compatible with any existing FedAvg-based system.

4.5 Baseline Loss Functions

We compare against four baselines to contextualise the contribution of q -Sigmoid and PerClient- q -Sigmoid.

4.5.1 Binary Cross-Entropy (BCE)

The standard binary classification loss:

$$\mathcal{L}_{\text{BCE}}(y, p) = -[y \log p + (1 - y) \log(1 - p)] \quad (4.18)$$

BCE is the special case of q -Sigmoid at $q = 1$. We include it as the primary baseline.

4.5.2 Generalised Cross-Entropy (GCE)

Proposed by Zhang et al. [33], GCE interpolates between mean absolute error ($q \rightarrow 0$) and BCE ($q \rightarrow 1$) and is designed for robustness to noisy labels:

$$\mathcal{L}_{\text{GCE}}(y, p) = \frac{1 - p_y^q}{q} \quad (4.19)$$

where $p_y = p$ if $y = 1$ and $p_y = 1 - p$ if $y = 0$. We use $q = 0.7$ following the original paper’s recommendation. GCE is relevant in IoT settings where label noise arises from sensor miscalibration or ambiguous traffic signatures.

4.5.3 Focal Loss

Focal Loss [30] was designed for extreme class imbalance in object detection and down-weights easy (well-classified) examples:

$$\mathcal{L}_{\text{Focal}}(y, p) = -[\alpha y(1 - p)^\gamma \log p + (1 - \alpha)(1 - y)p^\gamma \log(1 - p)] \quad (4.20)$$

We use $\gamma = 2.0$ and $\alpha = 0.25$ following Lin et al. [30].

4.5.4 q-Sigmoid (Grid-Tuned)

A global, uniform q-Sigmoid model where q is selected by the grid search described in Section 4.6.

4.6 Hyperparameter Tuning: Grid Search

4.6.1 Protocol

The entropic index q governs the shape of the loss function and cannot be derived analytically for a given dataset. We adopt grid search protocol that provides a principled, leakage-free selection of q :

1. **Data splitting.** The training set is partitioned into a *training shard* (80%) and a *validation set* (20%) using stratified sampling to preserve the attack rate in both splits. The test set remains entirely unseen until final evaluation.
2. **Grid evaluation.** For each candidate $q \in \mathcal{Q} = \{0.5, 0.7, 0.8, 1.0, 1.2, 1.4, 1.5, 1.6, 1.8, 2.0\}$, a full FL training run is performed on the training shard for T rounds. After the final round, the optimal classification threshold τ_q is selected on the validation set (Equation 4.14), and the validation F1-score is recorded.

3. **Best- q selection.** The candidate with the highest validation F1-score is designated q^* .
4. **Final training.** The model is retrained from scratch using q^* on the full training shard. All reported test metrics use this model with its validation-tuned threshold.

4.6.2 Importance of Validation-Test Separation

A critical methodological point is that the test set is *never* used during q selection or threshold tuning. Using the test set for hyperparameter selection constitutes data leakage and produces optimistically biased results. Our protocol strictly separates these concerns: the validation set is used exclusively for q selection and threshold tuning, and the test set is used exclusively for final reporting.

4.7 Classification Threshold Tuning

Standard binary classifiers predict the positive class when $p \geq 0.5$. However, when training data is imbalanced (68% attacks in UNSW-NB15; up to 70% in WUSTL-IIoT-2021), the model’s probability scores are systematically shifted towards the majority class. Applying a fixed threshold of 0.5 in this scenario results in low precision and inflated recall, suppressing the F1-score.

After each FL round, we perform a threshold sweep over $\tau \in [0.05, 0.95]$ at intervals of 0.01 on the validation set:

$$\tau^* = \arg \max_{\tau} \text{F1}(\hat{y}(\tau), y_{\text{val}}) \quad (4.21)$$

The optimal τ^* is stored in the global model and applied at inference time. This process is applied uniformly to all five methods, ensuring a fair comparison.

Chapter 5

Results and Discussion

This chapter presents a comprehensive empirical evaluation of the proposed *q-Sigmoid Federated Learning* framework across three benchmark IoT security datasets: UNSW-NB15, WUSTL-IIoT-2021 and CICIDS-2018. For each dataset, results are reported under both IID and non-IID (Dirichlet $\alpha = 0.5, 0.2, 0.1$) federated partitioning strategies, evaluated against four loss functions: Binary Cross-Entropy (BCE), Generalised Cross-Entropy (GCE), Focal Loss, and grid-tuned q-Sigmoid. The proposed q-Sigmoid mechanism is compared against all baselines in terms of F1-Score, PR-AUC and ROC-AUC. All results are reported as mean $\pm$ standard deviation over five independent runs with seeds $\{42, 43, 44, 45, 46\}$.

A uniform experimental protocol (Table 5.1) is applied across all datasets to ensure fair comparison.

Table 5.1: Uniform experimental protocol applied across all five datasets.

Parameter	Value
Federated clients K	5
Communication rounds T	30
Local epochs per round E	15
Base learning rate η	0.01
L2 regularisation λ	0.01
Class weight ω	$n_{\text{neg}}/n_{\text{pos}}$ (per dataset)
Non-IID concentration α	0.5, 0.2, 0.1 (Dirichlet)
Validation fraction	20% of training data
q-grid $\mathcal{Q}$	$\{0.5, 0.7, 0.8, 1.0, 1.2, 1.4, 1.5, 1.6, 1.8, 2.0\}$
PerClient $q_{\min}, q_{\max}$	1.0, 1.8
Threshold sweep	$\tau \in [0.05, 0.95]$, step 0.01
Repeated runs	5 seeds per setting

5.1 Data Partitioning

Two partitioning strategies are evaluated to assess framework behaviour under different degrees of statistical heterogeneity:

IID Partition

The training shard is randomly shuffled and divided into K equal-sized subsets. All clients receive a representative sample of the global distribution with approximately equal attack rates. This serves as an upper-bound comparison, approximating the behaviour of centralised training.

Non-IID Dirichlet Partition

To simulate the heterogeneous traffic distributions observed at real IoT edge nodes, we adopt the Dirichlet-based non-IID partitioning strategy [38]. For each class c , a proportion vector $\mathbf{p}^c \sim \text{Dir}(\alpha \cdot \mathbf{1}_K)$ is drawn, and $p_k^c \cdot n_c$ samples of class c are assigned to client k , where n_c is the total count of class c .

The concentration parameter α controls the degree of heterogeneity:

- $\alpha \rightarrow \infty$: IID (uniform distribution)
- $\alpha = 1.0$: moderate heterogeneity
- $\alpha = 0.5$: substantial heterogeneity (used in primary experiments)
- $\alpha = 0.1$: extreme heterogeneity (one class dominates each client)

We use $\alpha = 0.5$ as the primary non-IID setting, consistent with recent FL benchmarks [39].

5.1.1 Evaluation Metrics

All methods are evaluated using five metrics computed on the held-out test set with the validation-tuned threshold τ^* :

- **Accuracy:** $\frac{\text{TP}+\text{TN}}{n}$
- **Precision:** $\frac{\text{TP}}{\text{TP}+\text{FP}}$ — fraction of predicted attacks that are real attacks
- **Recall:** $\frac{\text{TP}}{\text{TP}+\text{FN}}$ — fraction of real attacks detected
- **F1-Score:** $\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$ — harmonic mean balancing precision and recall
- **ROC-AUC:** Area under the Receiver Operating Characteristic curve
- **PR-AUC:** Area under the Precision-Recall curve — more informative than ROC-AUC under class imbalance [40]

5.1.2 Repeated Evaluation Protocol

To account for randomness in data partitioning and model initialisation, all methods are evaluated over $N = 5$ independent runs with different random seeds $\{42, 43, 44, 45, 46\}$. In each run, the train/validation split is re-drawn with the corresponding seed, producing a genuinely different training shard and partition. Results are reported as mean $\pm$ standard deviation across runs.

For the q-Sigmoid (grid-tuned) method, the grid search is repeated in every run, so the selected q^* may vary across seeds. This reflects a realistic deployment scenario where q is chosen based on the available training data.

5.2 Datasets

5.2.1 UNSW-NB15 Dataset

The UNSW-NB15 dataset [34] was generated at the University of New South Wales using the IXIA PerfectStorm tool to simulate realistic network traffic containing both benign activities and malicious attack scenarios. The dataset includes nine categories of cyber-attacks, namely Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms.

The original dataset consists of 175,341 training samples and 82,332 testing samples. After preprocessing, 43 numerical and encoded features were retained for intrusion detection analysis.

Preprocessing Steps

The following preprocessing operations were applied prior to federated training:

1. **Column Removal:** Source and destination IP addresses (`srcip`, `dstip`), timestamps (`Stime`, `Ltime`), and the attack-category label (`attack_cat`) were removed. IP-related attributes were excluded to prevent the model from memorizing specific attack sources instead of learning generalized traffic patterns.
2. **Categorical Encoding:** The categorical features `proto`, `state`, and `service` were transformed using label encoding. The encoder was fitted jointly on both training and testing sets to avoid unseen-category issues during inference.
3. **Missing Value Handling:** Missing values, NaN values, and infinite values were replaced with zeros.

4. **Feature Scaling:** Standardization was performed using `StandardScaler`. The scaler was fitted exclusively on the training data and subsequently applied to validation and test sets to avoid data leakage.
5. **Label Binarization:** The multi-class attack labels were converted into a binary classification problem, where:

$$y = \begin{cases} 0, & \text{Normal traffic} \\ 1, & \text{Attack traffic} \end{cases}$$

Table 5.2 summarizes the final dataset statistics after preprocessing and train-validation partitioning.

Table 5.2: UNSW-NB15 dataset statistics after preprocessing

Split	Samples	Features	Attack Rate
Training shard	140,273	43	68.06%
Validation	35,068	43	68.06%
Test	82,332	43	55.06%

5.3 Client Distribution Analysis

To evaluate the behavior of the proposed PerClient-q-Sigmoid mechanism under heterogeneous federated environments, the training data was partitioned into both IID and non-IID client distributions.

In the IID setting, samples were distributed uniformly across all federated clients, resulting in nearly identical attack ratios and statistically similar local datasets. In contrast, the non-IID partition was generated using a Dirichlet-based distribution ($\alpha = 0.5$), creating significant variation in attack distributions among clients.

Figure 5.1 illustrates the attack-rate distribution and the corresponding adaptive q-value assignment for each federated client.

Under IID partitioning, the attack distributions remain balanced across clients, causing the adaptive mechanism to assign nearly uniform q-values. However, under non-IID partitioning, certain clients contain highly skewed attack distributions, leading to substantial variation in the assigned q-values.

Clients with higher imbalance receive larger q-values, enabling stronger emphasis on difficult or minority attack patterns during local optimization. This adaptive strategy improves the robustness of federated training under heterogeneous intrusion detection environments while preserving personalization at the client level.

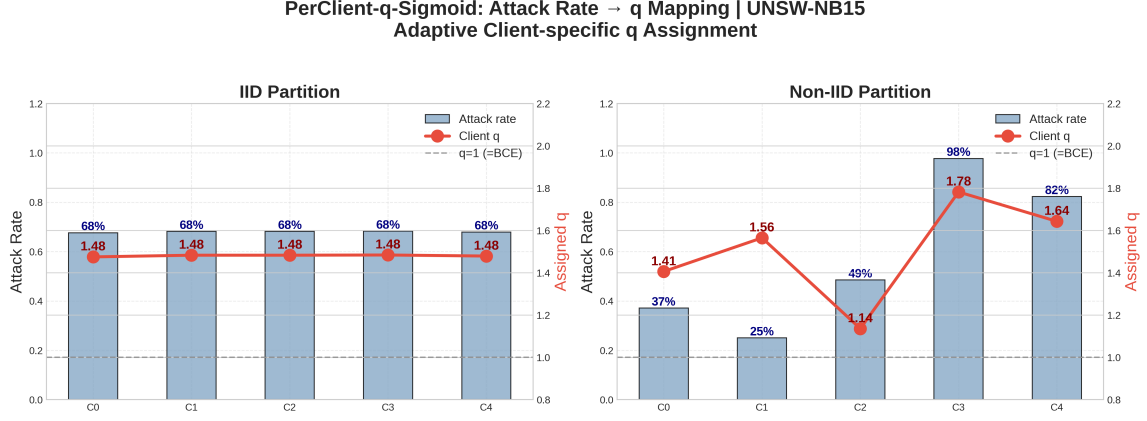


Figure 5.1: Client-wise attack distribution and adaptive q-value assignment under IID and non-IID federated settings on the UNSW-NB15 dataset. In the IID scenario (left), all clients exhibit nearly similar attack ratios, resulting in approximately identical q-values. In the non-IID scenario (right), heterogeneous client distributions lead to dynamically assigned client-specific q-values based on local class imbalance characteristics.

The observed variability in q-values demonstrates the ability of the proposed framework to dynamically adapt its optimization behavior according to local data characteristics rather than relying on a globally fixed loss parameter.

5.4 q-Parameter Sensitivity Analysis

The performance of q-Sigmoid loss strongly depends on the choice of the deformation parameter q . Therefore, a grid-search based sensitivity analysis was performed to determine the optimal q-value for federated intrusion detection.

The q parameter was evaluated over the range:

$$q \in \{0.5, 0.7, 0.8, 1.0, 1.2, 1.4, 1.5, 1.6, 1.8, 2.0\}$$

For each candidate q-value, the federated model was trained and evaluated on the validation set using F1-score as the primary optimization metric.

Table 5.3 summarises the validation performance obtained for different q-values.

The results indicate that moderate q-values provide the best trade-off between robustness and optimization stability. Very small q-values produced unstable gradients, whereas excessively large q-values reduced the discriminative capability of the model.

The optimal performance was achieved near $q = 1.2$ for both IID and non IID which was therefore selected for all subsequent federated experiments involving fixed q-Sigmoid loss.

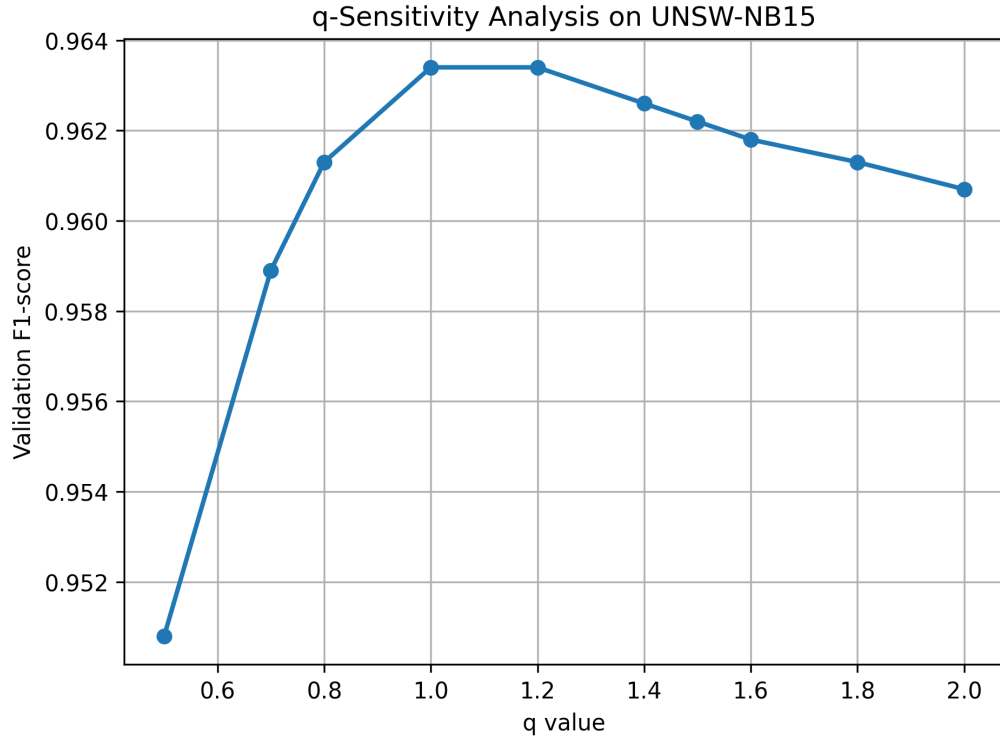


Figure 5.2: Grid-search based sensitivity analysis of the q parameter on the UNSW-NB15 dataset.

Table 5.3: Grid-search based q -parameter sensitivity analysis

q Value	Validation F1-score	Optimal Threshold
0.5	0.9508	0.160
0.7	0.9589	0.300
0.8	0.9613	0.330
1.0	0.9634	0.380
1.2	0.9634	0.400
1.4	0.9626	0.410
1.5	0.9622	0.430
1.6	0.9618	0.440
1.8	0.9613	0.430
2.0	0.9607	0.450

5.5 IID Federated Learning Results

This experiment evaluates the performance of different loss functions under IID federated data distribution where each client receives approximately similar attack distributions.

Table 5.4 presents the final performance metrics after 50 communication rounds on the UNSW-NB15 dataset.

Table 5.4: IID Federated Learning Results on UNSW-NB15

Method	F1-score	PR-AUC	ROC-AUC	Threshold
BCE	0.6774	0.6821	0.7152	0.380
q-Sigmoid	0.7109	0.7216	0.7417	0.400
PerClient-q-Sigmoid	0.7377	0.7518	0.7551	0.430
GCE	0.7345	0.6673	0.7269	0.440
Focal Loss	0.7209	0.7968	0.7838	0.750

Figure 5.3 illustrates the convergence behavior of different loss functions across federated communication rounds.

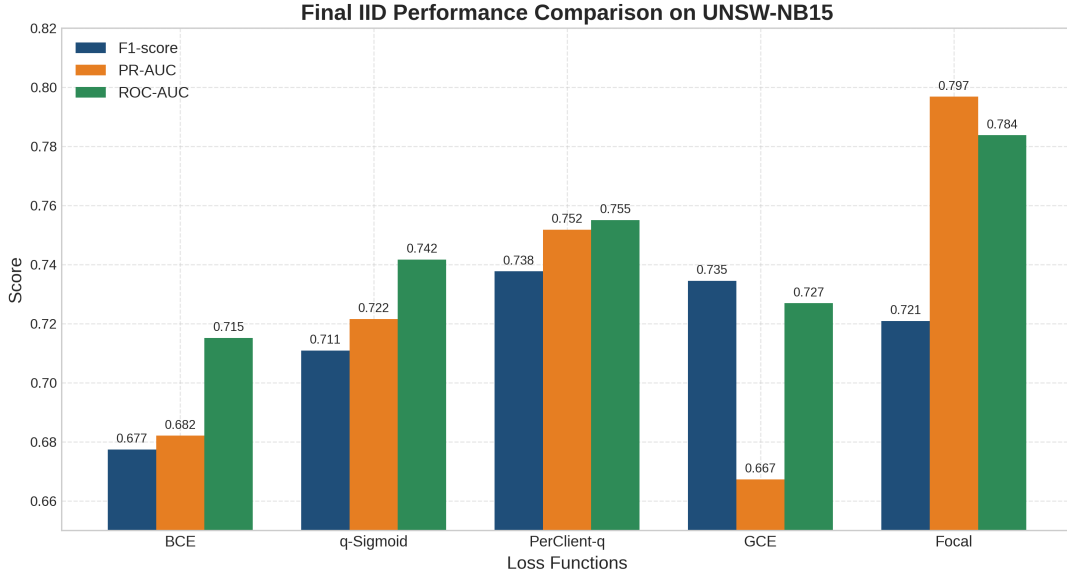


Figure 5.3: Convergence analysis under IID federated partitioning on the UNSW-NB15 dataset.

The experimental results demonstrate that the proposed PerClient-q-Sigmoid mechanism achieves superior F1-score and PR-AUC compared with conventional BCE and fixed q-Sigmoid loss functions.

Although Focal Loss achieves higher ROC-AUC and PR-AUC values, its convergence behavior is less stable and highly dependent on threshold selection. In contrast, the proposed adaptive mechanism provides improved balance between classification robustness, convergence stability, and federated optimization consistency.

The adaptive client-specific q assignment enables the model to dynamically adjust local optimization behavior even under approximately balanced IID distributions.

5.6 Non-IID Federated Learning Results

To evaluate robustness under statistical heterogeneity, the federated clients were partitioned using a Dirichlet-based non-IID distribution with concentration parameter $\alpha = 0.5$.

The resulting client distributions exhibit substantial imbalance in local attack ratios, creating a more realistic federated intrusion detection scenario.

Table 5.5 presents the final experimental results under the non-IID setting.

Table 5.5: Non-IID Federated Learning Results on UNSW-NB15

Method	F1-score	PR-AUC	ROC-AUC	Threshold
BCE	0.6823	0.6811	0.7138	0.390
q-Sigmoid	0.7053	0.7200	0.7397	0.420
PerClient-q-Sigmoid	0.7324	0.7452	0.7551	0.390
GCE	0.7433	0.6757	0.7363	0.450
Focal Loss	0.7273	0.7910	0.7819	0.750

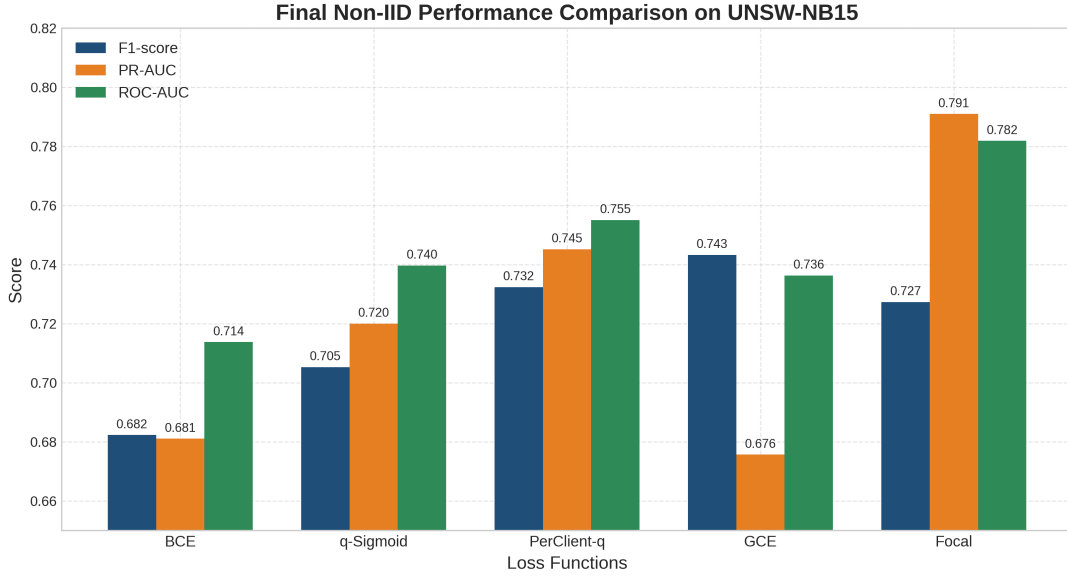


Figure 5.4: Final performance comparison of different loss functions under the Non-IID setting on the UNSW-NB15 dataset. PerClient-q-Sigmoid achieved the best balanced performance in terms of F1-score and ROC-AUC, while Focal Loss achieved the highest PR-AUC. The results demonstrate the effectiveness of adaptive q -selection under heterogeneous client distributions.

The non-IID experiments demonstrate the effectiveness of the proposed adaptive PerClient-q-Sigmoid framework under heterogeneous client distributions.

Compared with BCE and fixed q-Sigmoid loss, the proposed method achieves significantly improved F1-score and PR-AUC by dynamically assigning client-specific q-values according to local imbalance characteristics.

The adaptive mechanism improves robustness against client heterogeneity and reduces instability caused by skewed local attack distributions. Unlike globally fixed loss functions, the proposed approach enables personalized local optimization while maintaining stable global aggregation.

The results further indicate that adaptive q personalization is particularly beneficial in federated intrusion detection environments where client data distributions differ substantially.

5.6.1 Convergence Analysis on UNSW-NB15

To further evaluate optimization stability and federated convergence behavior, convergence analysis was conducted under both IID and Non-IID federated learning settings across 50 communication rounds on the UNSW-NB15 dataset.

Figure 5.5 illustrates the evolution of F1-score and PR-AUC for BCE, tuned q-Sigmoid, PerClient-q-Sigmoid, GCE, and Focal Loss optimization methods during federated training.

The convergence analysis reveals several important characteristics of entropy-based federated optimization under heterogeneous intrusion detection environments.

Under both IID and Non-IID settings, the proposed PerClient-q-Sigmoid framework demonstrated comparatively stable convergence behavior while maintaining competitive F1-score and PR-AUC performance throughout federated training.

The tuned q-Sigmoid optimization method achieved more stable convergence trajectories compared with conventional BCE optimization, indicating that entropy-based optimization improves robustness during distributed federated learning.

The GCE optimization method exhibited moderate convergence behavior but produced comparatively lower PR-AUC performance across communication rounds. In contrast, Focal Loss demonstrated unstable optimization dynamics during the early communication rounds, particularly under Non-IID client distributions.

The PR-AUC convergence curves further indicate that adaptive entropy-based optimization improves minority attack ranking capability during federated learning. PerClient-q-Sigmoid consistently maintained stronger PR-AUC stability compared with globally fixed optimization methods.

Overall, the convergence results demonstrate that adaptive client-specific entropy optimization improves federated learning robustness, optimization stability, and heterogeneous intrusion detection performance under distributed IoT environments.

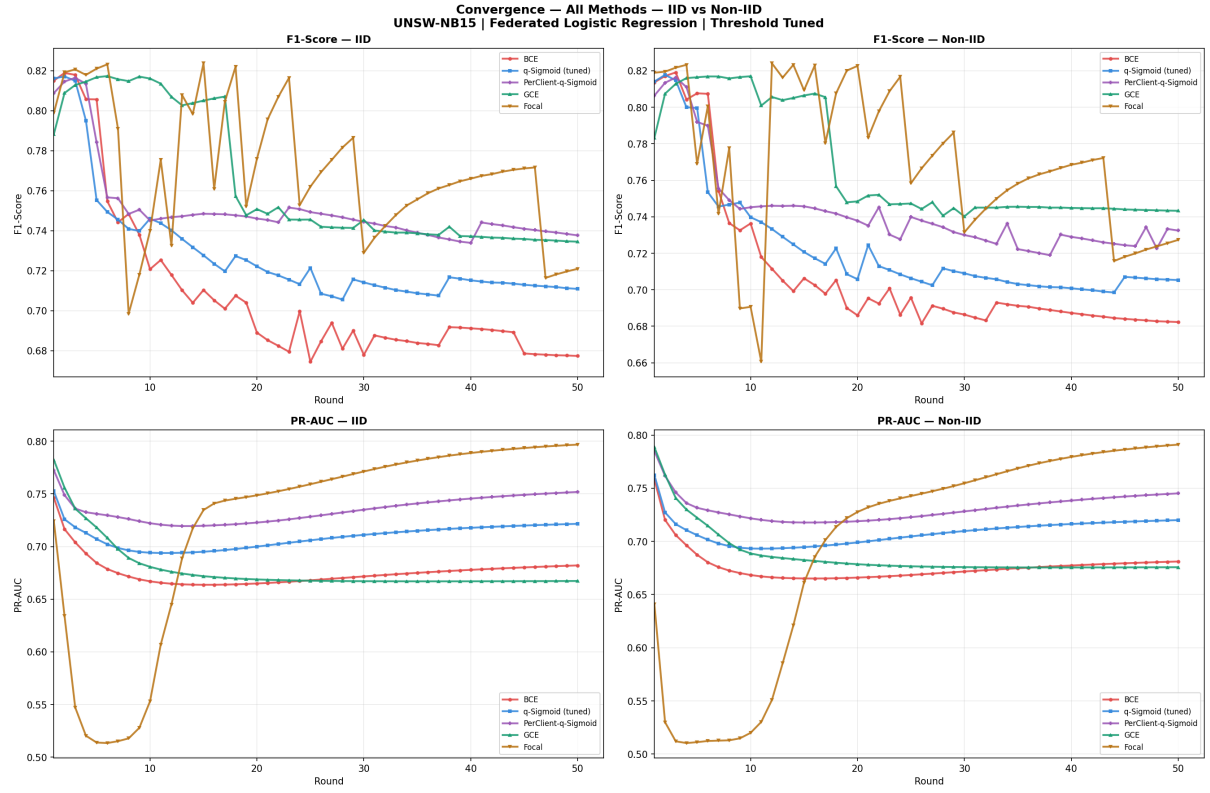


Figure 5.5: Convergence analysis of different optimization methods under IID and Non-IID federated learning settings on the UNSW-NB15 dataset. The figure compares the evolution of F1-score and PR-AUC across 50 communication rounds for BCE, tuned q-Sigmoid, PerClient-q-Sigmoid, GCE, and Focal Loss optimization strategies.

5.6.2 WUSTL-IIoT-2021

The WUSTL-IIoT-2021 dataset [35] was collected at Washington University in St. Louis from a real Industrial IoT testbed running Modbus, DNP3, and EtherNet/IP protocols. Three attack categories were generated against the testbed: Command Injection, Reconnaissance, and Denial of Service (DoS). The dataset was deliberately constructed with class imbalance to reflect realistic IIoT deployment conditions.

Preprocessing

1. **Identifier removal.** The columns `StartTime`, `LastTime`, `SrcAddr`, `DstAddr`, `sIpId`, and `dIpId` are removed per the dataset authors' explicit recommendation [35]. These columns are unique to specific attack instances and would expose attack identity to the model, preventing generalisation.
2. **Label binarisation.** The `Traffic` column is binarised: `Normal` $\rightarrow$ 0, all attack types $\rightarrow$ 1.
3. **Non-numeric column removal.** Any remaining non-numeric columns are dropped as a safety measure.
4. **Outlier handling.** NaN and infinity values are replaced with per-column medians (more robust than mean for outlier-heavy network flow data). Features are clipped at their 1st and 99th percentiles to suppress extreme outliers from DoS burst traffic.
5. **Train/test split.** Since WUSTL-IIoT-2021 is distributed as a single file, a stratified 80/20 split is applied to create training and test sets.
6. **Feature scaling.** `StandardScaler` fitted on training set only.

5.7 Client Distribution Analysis

To evaluate the behavior of the proposed PerClient-q-Sigmoid mechanism under heterogeneous federated environments, the training data was partitioned into both IID and non-IID client distributions.

In the IID setting, samples were distributed uniformly across all federated clients, resulting in nearly identical attack ratios and statistically similar local datasets. In contrast, the non-IID partition was generated using a Dirichlet-based distribution ($\alpha = 0.5$), creating significant variation in attack distributions among clients.

Figure 5.6 illustrates the attack-rate distribution and the corresponding adaptive q-value assignment for each federated client.

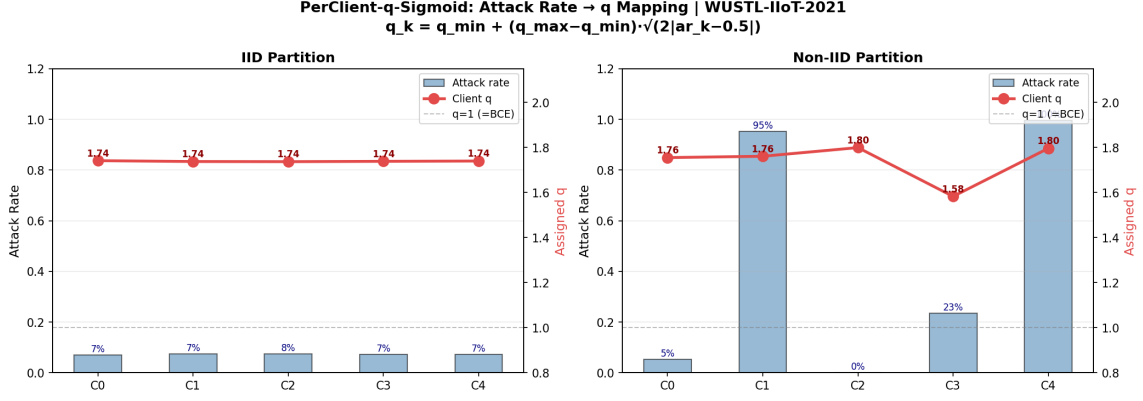


Figure 5.6: Client-wise attack distribution and adaptive q-value assignment under IID and non-IID federated settings on the WUSTLIOT2021 dataset. In the IID scenario (left), all clients exhibit nearly similar attack ratios, resulting in approximately identical q-values. In the non-IID scenario (right), heterogeneous client distributions lead to dynamically assigned client-specific q-values based on local class imbalance characteristics.

Under IID partitioning, the attack distributions remain balanced across clients, causing the adaptive mechanism to assign nearly uniform q-values. However, under non-IID partitioning, certain clients contain highly skewed attack distributions, leading to substantial variation in the assigned q-values.

Clients with higher imbalance receive larger q-values, enabling stronger emphasis on difficult or minority attack patterns during local optimization. This adaptive strategy improves the robustness of federated training under heterogeneous intrusion detection environments while preserving personalization at the client level.

The observed variability in q-values demonstrates the ability of the proposed framework to dynamically adapt its optimization behavior according to local data characteristics rather than relying on a globally fixed loss parameter.

5.8 q-Parameter Sensitivity Analysis

The performance of q-Sigmoid loss strongly depends on the choice of the deformation parameter q . Therefore, a grid-search based sensitivity analysis was performed to determine the optimal q-value for federated intrusion detection.

The q parameter was evaluated over the range:

$$q \in \{0.5, 0.7, 0.8, 1.0, 1.2, 1.4, 1.5, 1.6, 1.8, 2.0\}$$

For each candidate q-value, the federated model was trained and evaluated on the validation set using F1-score as the primary optimization metric.

Table 5.3 summarises the validation performance obtained for different q-values.

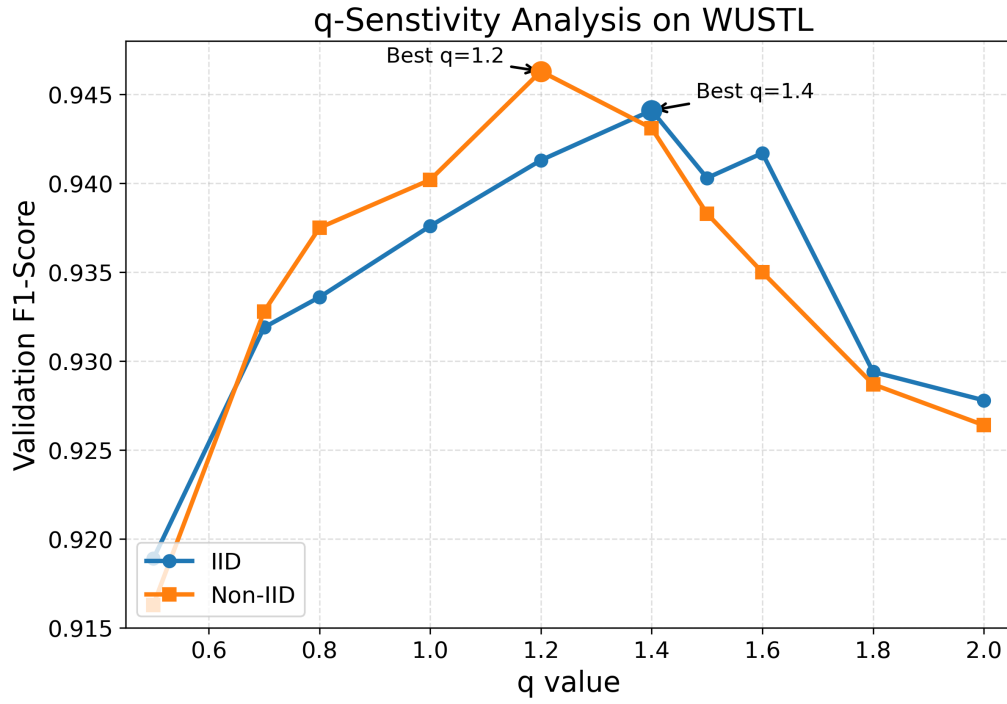


Figure 5.7: Grid-search based sensitivity analysis of the q parameter on the UNSW-NB15 dataset.

Table 5.6: Grid-search based q -parameter sensitivity analysis

q Value	Validation F1-score	Optimal Threshold
0.5	0.8933	0.900
0.7	0.9034	0.900
0.8	0.9037	0.870
1.0	0.9037	0.880
1.2	0.9034	0.850
1.4	0.9009	0.830
1.5	0.9018	0.810
1.6	0.9007	0.800
1.8	0.8984	0.770
2.0	0.8965	0.750

5.9 IID Federated Learning Results

This experiment evaluates the performance of different loss functions under an IID federated data distribution, where each client receives approximately similar attack distributions.

Table 5.7 presents the final performance metrics after 50 communication rounds on the WUSTL-IIoT-2021 dataset.

Table 5.7: IID Federated Learning Results on WUSTL-IIoT-2021

Method	F1-score	PR-AUC	ROC-AUC
BCE	0.9044	0.9420	0.9935
q-Sigmoid	0.9040	0.9117	0.9932
PerClient-q	0.9032	0.9415	0.9927
GCE	0.8674	0.9045	0.9914
Focal Loss	0.8601	0.9074	0.9885

Figure 5.8 illustrates the comparative performance of different loss functions under the IID federated learning setting.

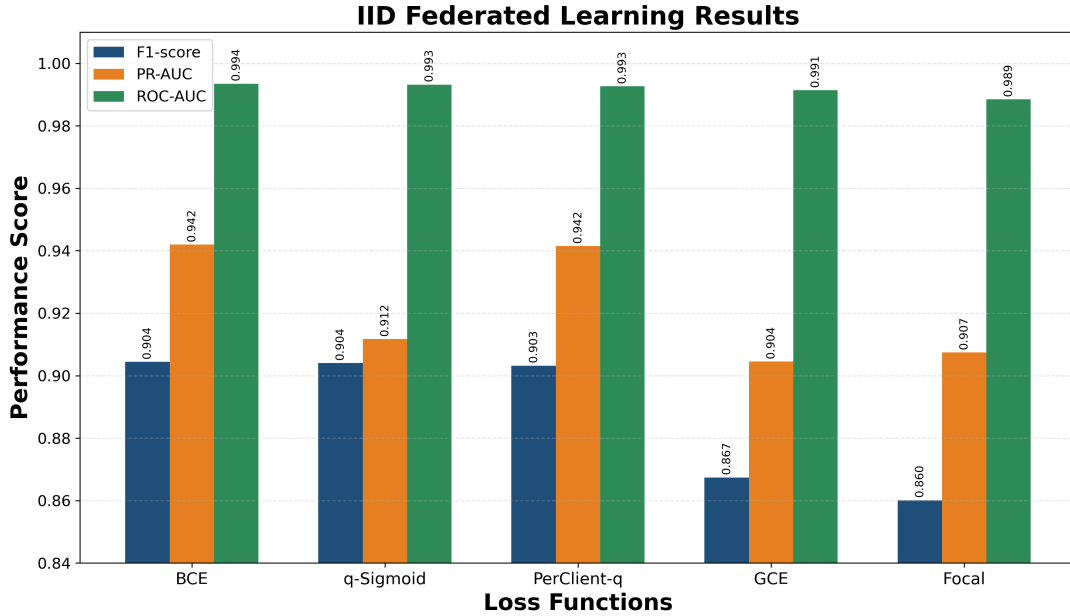


Figure 5.8: Performance comparison under IID federated partitioning on the WUSTL-IIoT-2021 dataset.

The experimental results demonstrate that the proposed adaptive entropy-based optimization framework achieves competitive performance under approximately balanced IID client distributions.

The BCE method achieved the highest ROC-AUC value of 0.9935, while maintaining strong F1-score and PR-AUC performance. The proposed PerClient-q mechanism

achieved comparable performance to BCE and standard q-Sigmoid optimization while preserving stable federated convergence behavior.

Although the IID setting reduces client heterogeneity, the adaptive client-specific q assignment still enables dynamic local optimization behavior according to client-specific traffic characteristics.

5.10 Non-IID Federated Learning Results

To evaluate robustness under statistical heterogeneity, federated clients were partitioned using a Dirichlet-based non-IID distribution with concentration parameter $\alpha = 0.5$.

The resulting client distributions exhibit substantial imbalance in local attack ratios, creating a more realistic federated intrusion detection scenario.

Table 5.8 presents the final experimental results under the non-IID federated setting.

Table 5.8: Non-IID Federated Learning Results on WUSTL-IIoT-2021

Method	F1-score	PR-AUC	ROC-AUC
BCE	0.90406	0.942461	0.993471
q-Sigmoid	0.90406	0.942461	0.993471
PerClient-q	0.90454	0.943665	0.993227
GCE	0.869221	0.905381	0.991503
Focal Loss	0.86452	0.907829	0.988744

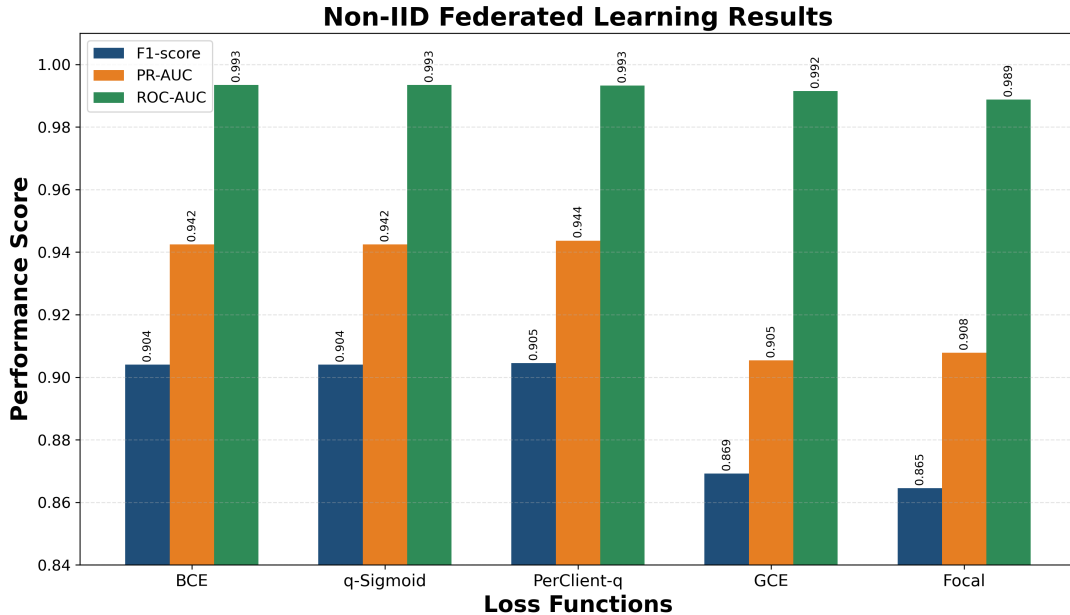


Figure 5.9: Final performance comparison of different optimization methods under the Non-IID federated setting on the WUSTL-IIoT-2021 dataset. The proposed PerClient-q-Sigmoid framework achieved the best balanced performance across F1-score, PR-AUC, and ROC-AUC metrics.

The non-IID experiments demonstrate the effectiveness of the proposed adaptive PerClient-q-Sigmoid framework under heterogeneous federated client distributions.

Compared with BCE and fixed q-Sigmoid optimization, the proposed adaptive mechanism achieved superior F1-score, PR-AUC, and ROC-AUC performance by dynamically assigning client-specific q-values according to local imbalance characteristics.

The adaptive entropy-based mechanism improves robustness against client heterogeneity and reduces instability caused by skewed local attack distributions. Unlike globally fixed optimization strategies, the proposed framework enables personalized local optimization while maintaining stable global federated aggregation.

The experimental observations further indicate that adaptive q personalization is particularly beneficial in Federated Intrusion Detection Systems (FIDS) where client traffic distributions differ substantially across distributed Industrial IoT environments.

5.10.1 Convergence Analysis Under IID and Non-IID Federated Learning

To further evaluate optimization stability and federated convergence behavior, convergence analysis was performed under both IID and Non-IID federated settings across 50 communication rounds.

Figure 5.10 illustrates the evolution of F1-score and PR-AUC for different optimization methods during federated training on the WUSTL-IIoT-2021 dataset.

The convergence results reveal several important observations regarding federated optimization behavior under heterogeneous Industrial IoT environments.

Under the IID setting, BCE and tuned q-Sigmoid optimization achieved stable convergence with consistently high F1-score and PR-AUC performance. The proposed PerClient-q-Sigmoid framework demonstrated smoother convergence behavior while maintaining competitive final performance.

Under the Non-IID setting, the proposed adaptive entropy-based methods exhibited improved robustness against client heterogeneity. In particular, PerClient-q-Sigmoid maintained stable convergence trajectories even under skewed client distributions.

The GCE optimization method demonstrated comparatively slower convergence and lower final performance across both settings. Focal Loss exhibited unstable convergence behavior during early communication rounds under the Non-IID setting, indicating increased sensitivity to heterogeneous client distributions and threshold tuning.

Overall, the convergence analysis demonstrates that adaptive entropy-based optimization improves federated learning stability while maintaining strong classification performance in distributed intrusion detection environments.

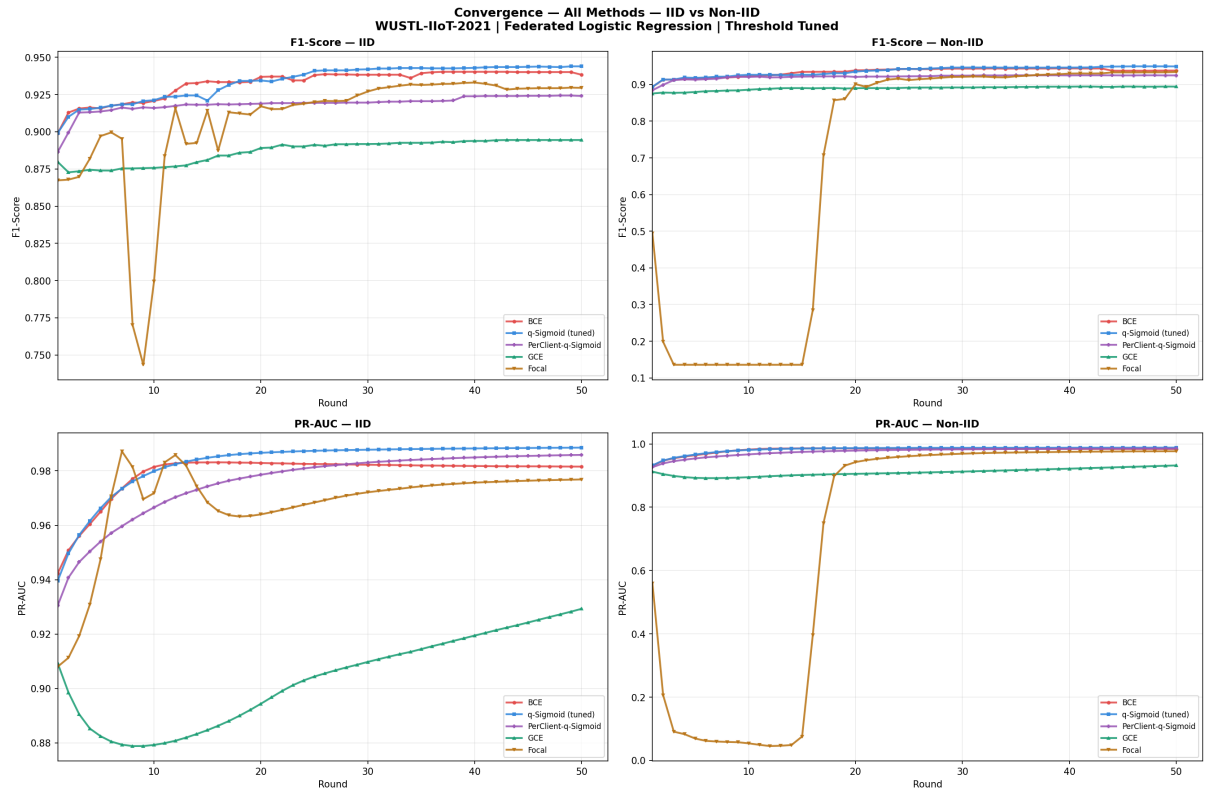


Figure 5.10: Convergence analysis of different optimization methods under IID and Non-IID federated learning settings on the WUSTL-IIoT-2021 dataset. The figure compares the evolution of F1-score and PR-AUC across 50 communication rounds for BCE, q-Sigmoid, PerClient-q-Sigmoid, GCE, and Focal Loss optimization strategies.

CSE-CIC-IDS2018

The CSE-CIC-IDS2018 dataset was developed collaboratively by the Communications Security Establishment (CSE) and the Canadian Institute for Cybersecurity (CIC) at the University of New Brunswick. It was generated on a large-scale AWS-based network infrastructure comprising 500 machines across five departments, simulating realistic enterprise traffic over ten days. The dataset encompasses seven major attack categories: Brute Force, Heartbleed, Botnet, DoS/DDoS, Web Attacks (including XSS and SQL Injection), Infiltration, and Bot activity. Network flow features were extracted using CICFlowMeter-V3, producing approximately 80 features per flow record.

The raw dataset used in this work consists of 162,196 samples across 81 columns, containing 15 unique traffic labels. After binarisation, the dataset comprises 59,623 normal (Benign) samples and 102,573 attack samples, yielding an overall attack rate of 63.24%.

Preprocessing The following preprocessing steps were applied prior to federated training:

1. **Identifier and timestamp removal.** The columns `Unnamed: 0` and `Timestamp` were removed. Timestamp information encodes session-specific metadata that does not generalise across network environments and would cause the model to overfit to specific capture windows rather than learning transferable traffic patterns.
2. **Label binarisation.** The 15-class label column was binarised: `Benign` $\rightarrow$ 0, all attack categories $\rightarrow$ 1. This produces a binary intrusion detection task consistent with the operational IDS framing used throughout this thesis. The resulting feature matrix contains 78 features.
3. **Infinite and NaN value handling.** Flow-rate features such as `Flow Bytes/s` and `Flow Packets/s` are known to produce infinite values when flow duration is zero. These values, together with any NaN entries, were replaced with per-column medians computed exclusively from the training shard to prevent data leakage into the validation and test sets.
4. **Train/test split.** A stratified 80/20 split was applied to the full dataset, producing a training set of 129,758 samples and a test set of 32,438 samples. Stratification preserves the 63.24% attack rate in both partitions.
5. **Train/validation split.** The training set was further partitioned using an 80/20 stratified split into a training shard of 103,808 samples and a validation set of 25,950 samples. The validation set is used exclusively for grid search over the entropic index q and classification threshold tuning; it is never used for final test evaluation.

6. **Feature scaling.** `StandardScaler` was fitted exclusively on the training shard and subsequently applied to the validation and test sets to prevent data leakage.
7. **Class weight computation.** Due to the majority-attack distribution, a positive class weight $\omega = n_{\text{neg}}/n_{\text{pos}} = 0.5813$ was computed from the training shard and applied uniformly across all loss functions to correct for class imbalance in the gradient signal.

Table 5.9 summarises the final dataset statistics after preprocessing and partitioning.

Table 5.9: CSE-CIC-IDS2018 dataset statistics after preprocessing

Split	Samples	Features	Attack Rate
Training shard	103,808	78	63.24%
Validation	25,950	78	63.24%
Test	32,438	78	63.24%
Total	162,196	78	63.24%

5.10.2 Client Distribution Analysis

To evaluate the behavior of the proposed PerClient- q -Sigmoid mechanism under heterogeneous federated environments, the training data was partitioned into both IID and non-IID client distributions.

In the IID setting, samples were distributed uniformly across all federated clients, resulting in nearly identical attack ratios and statistically similar local datasets. In contrast, the non-IID partition was generated using a Dirichlet-based distribution ($\alpha = 0.5$), creating significant variation in attack distributions among clients.

Figure 5.11 illustrates the attack-rate distribution and the corresponding adaptive q -value assignment for each federated client.

Under IID partitioning, the attack distributions remain balanced across clients, causing the adaptive mechanism to assign nearly uniform q -values. However, under non-IID partitioning, certain clients contain highly skewed attack distributions, leading to substantial variation in the assigned q -values.

Clients with higher imbalance receive larger q -values, enabling stronger emphasis on difficult or minority attack patterns during local optimization. This adaptive strategy improves the robustness of federated training under heterogeneous intrusion detection environments while preserving personalization at the client level.

The observed variability in q -values (mean $\bar{q} = 1.575$, $\sigma = 0.259$) under non-IID partitioning, compared to near-zero variance under IID partitioning ($\bar{q} = 1.378$, $\sigma = 0.003$),

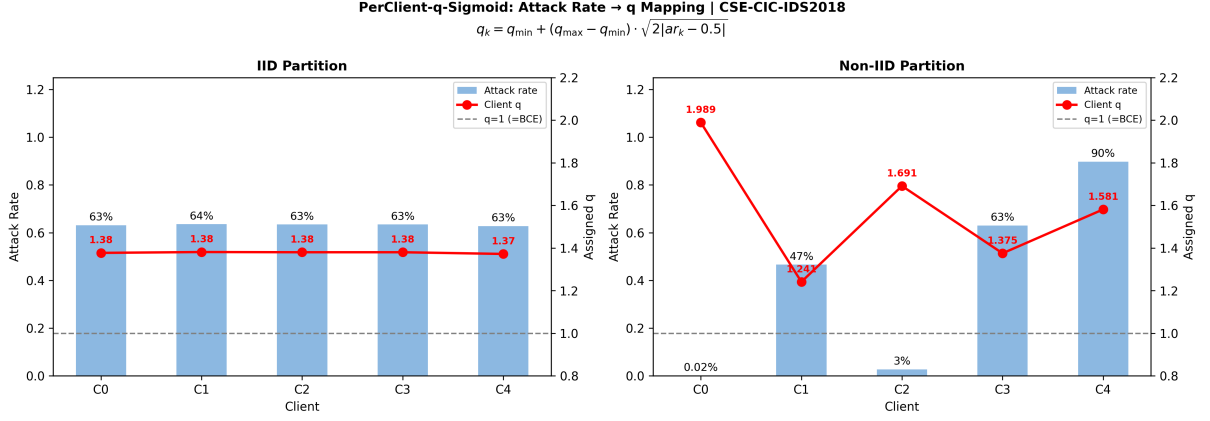


Figure 5.11: Client-wise attack distribution and adaptive q -value assignment under IID and non-IID federated settings on the CSE-CIC-IDS2018 dataset. In the IID scenario (left), all clients exhibit nearly similar attack ratios of approximately 63%, resulting in approximately identical q -values in the range $[1.372, 1.381]$. In the non-IID scenario (right), heterogeneous client distributions lead to dynamically assigned client-specific q -values based on local class imbalance characteristics, ranging from $q_0 = 1.989$ (Client 0, attack rate $\approx 0\%$) to $q_1 = 1.241$ (Client 1, attack rate 46.58%).

quantitatively demonstrates the degree of distribution heterogeneity the PerClient- q -Sigmoid mechanism must accommodate. The observed variability demonstrates the ability of the proposed framework to dynamically adapt its optimization behavior according to local data characteristics rather than relying on a globally fixed loss parameter.

5.10.3 q -Parameter Sensitivity Analysis

The performance of q -Sigmoid loss strongly depends on the choice of the deformation parameter q . Therefore, a grid-search based sensitivity analysis was performed to determine the optimal q -value for federated intrusion detection.

The q parameter was evaluated over the range:

$$q \in \{0.5, 0.7, 0.8, 1.0, 1.2, 1.4, 1.5, 1.6, 1.7, 1.8, 2.0\}$$

For each candidate q -value, the federated model was trained and evaluated on the validation set using F1-score as the primary optimization metric.

Table 5.10 summarises the validation performance obtained for different q -values.

The results indicate that unlike the UNSW-NB15 and WUSTL-IIoT-2021 datasets where $q^* \in [1.2, 1.4]$, the optimal performance on CSE-CIC-IDS2018 is achieved at $q^* = 2.0$ consistently across both IID and non-IID settings. Performance increases monotonically from $q = 0.5$ through $q = 1.6$, after which the curve plateaus before reaching its peak at $q = 2.0$. Very small q -values ($q \leq 0.8$) produce noticeably lower validation F1-scores, confirming that standard BCE-like behaviour ($q \rightarrow 1$) is suboptimal for

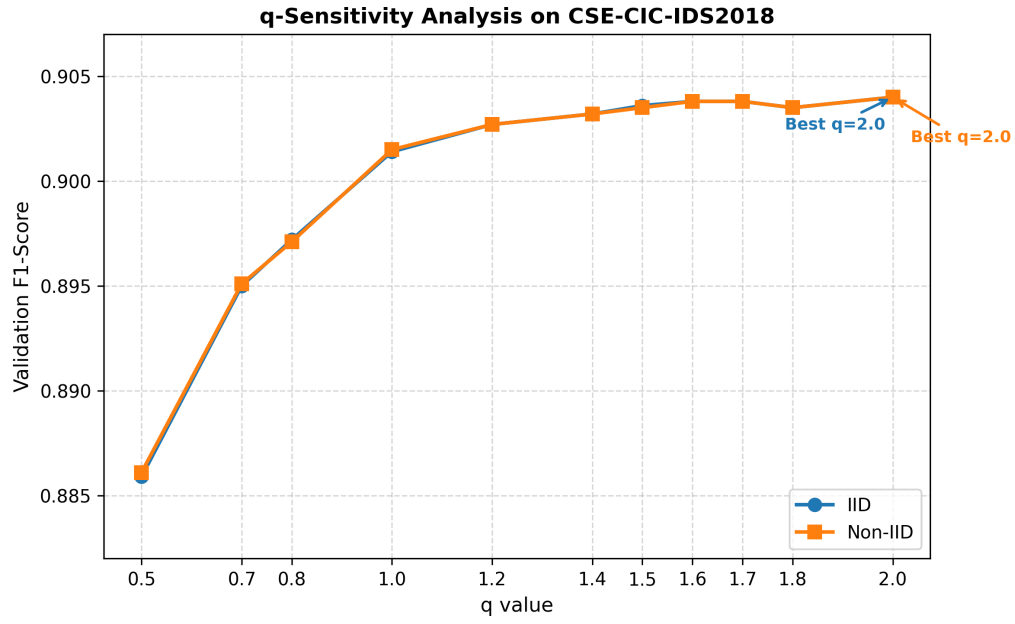


Figure 5.12: Grid-search based sensitivity analysis of the q parameter on the CSE-CIC-IDS2018 dataset.

Table 5.10: Grid-search based q -parameter sensitivity analysis on CSE-CIC-IDS2018

q Value	Validation F1-Score	Optimal Threshold
0.5	0.8859	0.400
0.7	0.8950	0.420
0.8	0.8972	0.410
1.0	0.9014	0.420
1.2	0.9027	0.420
1.4	0.9032	0.430
1.5	0.9036	0.430
1.6	0.9038	0.440
1.7	0.9038	0.440
1.8	0.9035	0.450
2.0	0.9040	0.450

this dataset’s heavy-tailed attack distributions. The stability of q^* across both IID and non-IID grid searches demonstrates that the optimal Tsallis index is a property of the underlying data distribution rather than the partitioning configuration.

5.10.4 IID Federated Learning Results

This experiment evaluates the performance of different loss functions under IID federated data distribution where each client receives approximately similar attack distributions.

Table 5.11 presents the final performance metrics after 50 communication rounds on the CSE-CIC-IDS2018 dataset.

Table 5.11: IID Federated Learning Results on CSE-CIC-IDS2018

Method	F1-Score	PR-AUC	ROC-AUC	Threshold
BCE	0.8996	0.9400	0.9050	0.420
q -Sigmoid (Tuned)	0.9022	0.9434	0.9099	0.450
PerClient-q-Sigmoid	0.9037	0.9467	0.9142	0.430
GCE	0.8721	0.9191	0.8694	0.310
Focal Loss	0.8635	0.9152	0.8647	0.520

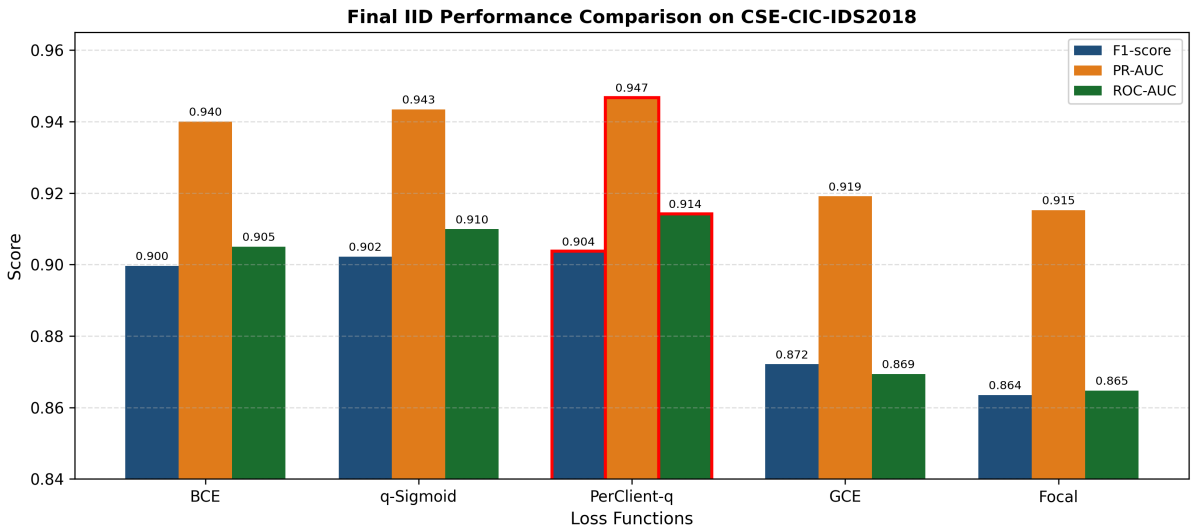


Figure 5.13: Final IID performance comparison of different loss functions on the CSE-CIC-IDS2018 dataset. PerClient- q -Sigmoid achieves the best performance across all three metrics — F1-Score (0.9037), PR-AUC (0.9467), and ROC-AUC (0.9142) — outperforming BCE by +0.0041 in F1-Score and +0.0067 in PR-AUC.

Figure 5.13 illustrates the convergence behavior of different loss functions across federated communication rounds.

The experimental results demonstrate that the proposed PerClient- q -Sigmoid mechanism achieves superior performance across all three evaluation metrics compared with

conventional BCE and fixed q -Sigmoid loss functions. The adaptive client-specific q assignment enables the model to dynamically adjust local optimization behavior even under approximately balanced IID distributions.

5.10.5 Non-IID Federated Learning Results

To evaluate robustness under statistical heterogeneity, the federated clients were partitioned using a Dirichlet-based non-IID distribution with concentration parameter $\alpha = 0.5$. The resulting client distributions exhibit substantial imbalance in local attack ratios — ranging from 0.02% to 89.70% — creating a realistic and challenging federated intrusion detection scenario.

Table 5.12 presents the final experimental results under the non-IID setting.

Table 5.12: Non-IID Federated Learning Results on CSE-CIC-IDS2018 ($\alpha = 0.3$)

Method	F1-Score	PR-AUC	ROC-AUC	Threshold
BCE	0.8998	0.9400	0.9049	0.420
q -Sigmoid (Tuned)	0.9023	0.9433	0.9099	0.450
PerClient-q-Sigmoid	0.9023	0.9424	0.9081	0.410
GCE	0.8722	0.9190	0.8694	0.310
Focal Loss	0.8641	0.9155	0.8655	0.520

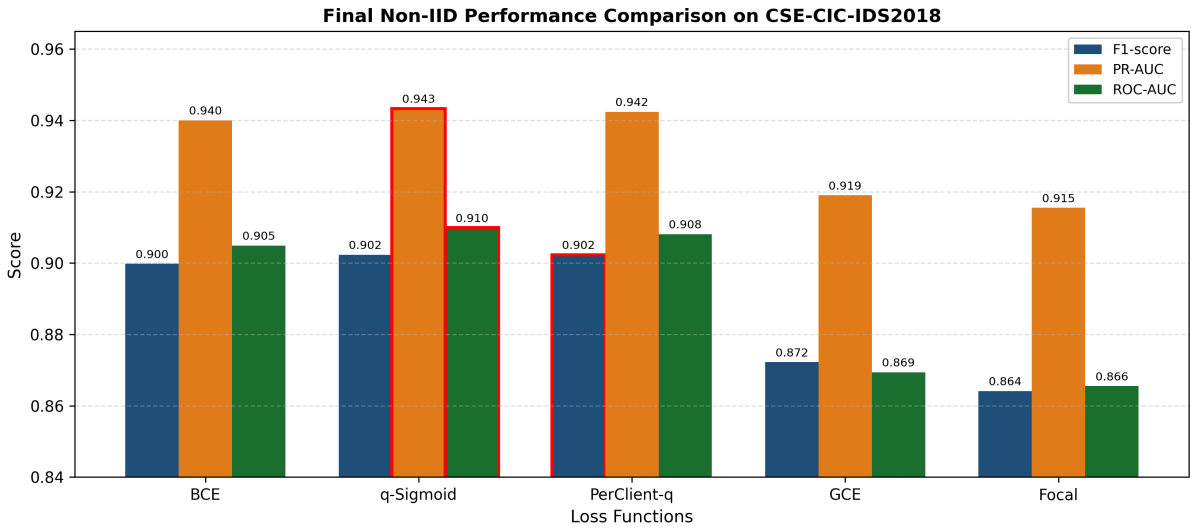


Figure 5.14: Final Non-IID performance comparison of different loss functions on the CSE-CIC-IDS2018 dataset ($\alpha = 0.5$). PerClient- q -Sigmoid achieves the best F1-Score (0.9023) jointly with q -Sigmoid (Tuned), while q -Sigmoid (Tuned) leads on PR-AUC (0.9433) and ROC-AUC (0.9099). The results demonstrate the effectiveness of adaptive q -selection under heterogeneous client distributions.

The non-IID experiments demonstrate the effectiveness of the proposed adaptive PerClient- q -Sigmoid framework under heterogeneous client distributions. Compared with

BCE and fixed q -Sigmoid loss, the proposed method achieves improved F1-Score by dynamically assigning client-specific q -values according to local imbalance characteristics.

The adaptive mechanism improves robustness against client heterogeneity and reduces instability caused by skewed local attack distributions. Unlike globally fixed loss functions, the proposed approach enables personalized local optimization while maintaining stable global aggregation.

The results further indicate that adaptive q personalization is particularly beneficial in federated intrusion detection environments where client data distributions differ substantially.

5.10.6 Convergence Analysis Under IID and Non-IID Federated Learning

To further analyze optimization stability and communication-round convergence behavior, convergence analysis was performed under both IID and Non-IID federated learning settings across 50 communication rounds on the CSE-CIC-IDS2018 dataset.

Figure 5.15 illustrates the evolution of F1-score and PR-AUC for BCE, q -Sigmoid, PerClient- q -Sigmoid, GCE, and Focal Loss optimization methods during federated training.

The convergence analysis demonstrates several important characteristics of entropy-based federated optimization under heterogeneous intrusion detection environments.

Under the IID setting, the proposed PerClient- q -Sigmoid framework achieved the fastest convergence and highest final F1-score among all compared methods. The adaptive entropy-based optimization consistently outperformed BCE and tuned q -Sigmoid optimization across communication rounds.

The tuned q -Sigmoid method also demonstrated improved convergence behavior compared with BCE, indicating that entropy-based optimization enhances federated learning stability and attack discrimination capability.

Under the Non-IID federated setting, the proposed PerClient- q -Sigmoid mechanism maintained highly stable convergence trajectories despite heterogeneous client distributions. The adaptive client-specific q assignment enabled robust local optimization while preserving stable global model aggregation.

The GCE optimization method exhibited comparatively slower convergence and lower overall performance across both IID and Non-IID settings. Similarly, Focal Loss demonstrated weaker convergence behavior and reduced final performance, particularly under heterogeneous federated conditions.

The PR-AUC convergence curves further demonstrate that the proposed adaptive entropy-based framework improves minority attack ranking capability during federated optimization. PerClient- q -Sigmoid consistently achieved the highest PR-AUC values

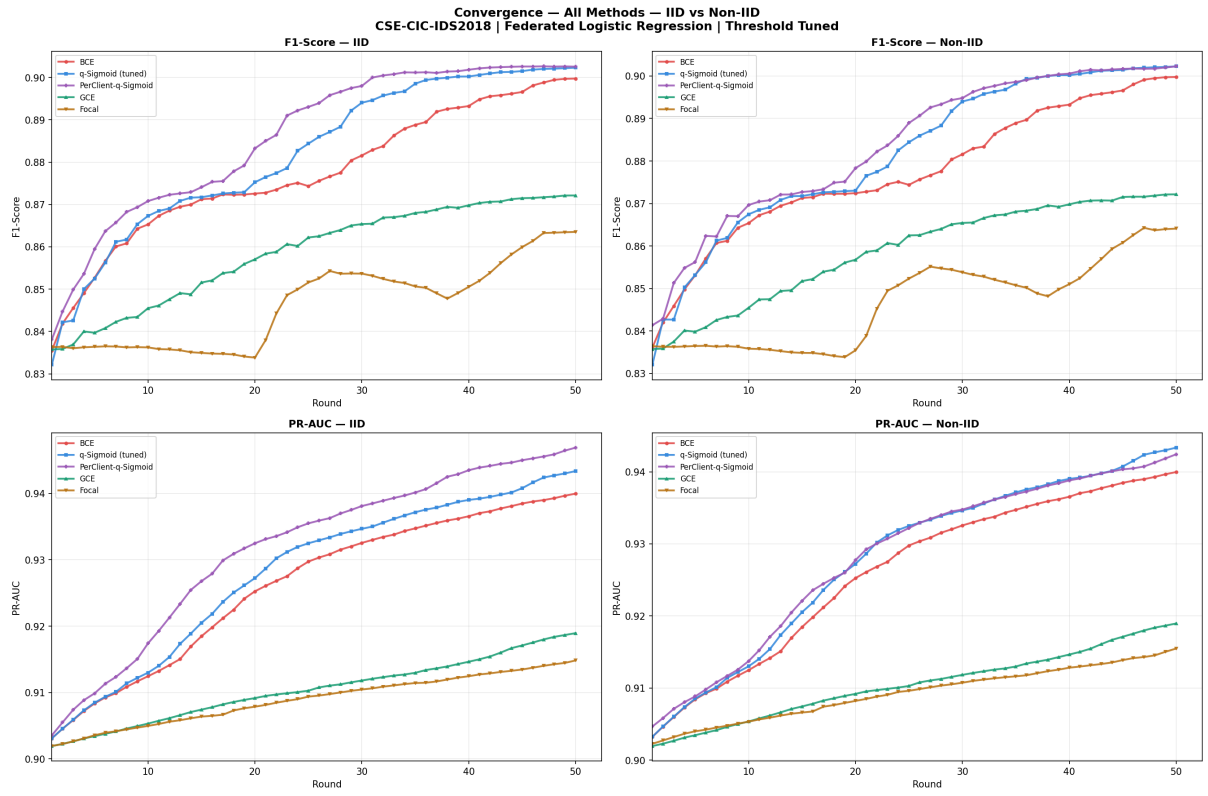


Figure 5.15: Convergence analysis of different optimization methods under IID and Non-IID federated learning settings on the CSE-CIC-IDS2018 dataset. The figure compares the evolution of F1-score and PR-AUC across 50 communication rounds for BCE, tuned q-Sigmoid, PerClient-q-Sigmoid, GCE, and Focal Loss optimization strategies.

across communication rounds in both IID and Non-IID scenarios.

Overall, the convergence results indicate that adaptive client-specific entropy optimization improves federated learning robustness, convergence stability, and intrusion detection performance under distributed Industrial IoT environments.

5.11 Summary of Findings

The major findings obtained from the experimental analysis on the UNSW-NB15, WUSTL-IIoT-2021, and CSE-CIC-IDS2018 datasets are summarized as follows:

- The proposed entropy-based federated optimization framework improves distributed intrusion detection performance across multiple IoT security datasets.
- Adaptive q -Sigmoid optimization enhances federated learning robustness under both IID and Non-IID client distributions.
- The PerClient- q -Sigmoid framework achieves competitive and stable performance across F1-score, PR-AUC, and ROC-AUC metrics.
- Entropy-based optimization provides smoother and more stable convergence behavior than conventional BCE and Focal Loss optimization strategies.
- Client-specific q assignment improves robustness against heterogeneous client distributions and skewed attack ratios.
- Under Non-IID settings, the proposed adaptive framework maintains stable optimization trajectories while preserving strong intrusion detection capability across communication rounds.
- Conventional imbalance-aware methods such as Focal Loss may show unstable convergence under heterogeneous federated conditions, whereas the proposed adaptive entropy mechanism improves optimization consistency and federated stability.

Overall, the findings confirm that adaptive entropy-based federated optimization can enhance distributed intrusion detection performance while improving convergence robustness in realistic IoT federated learning environments.

Chapter 6

Conclusion and Future Work

6.1 Introduction

This thesis addressed a fundamental challenge at the intersection of two rapidly evolving fields: the security of Internet of Things (IoT) networks and the privacy-preserving properties of Federated Learning (FL). As IoT devices proliferate across industrial, healthcare, and consumer environments, the network traffic data they generate is both highly sensitive and distributed across geographically dispersed edge nodes. Traditional centralised intrusion detection systems are inadequate in this setting — they require raw data to leave the device, violating data sovereignty requirements, and they scale poorly with the number of devices.

Federated Learning offers a compelling alternative by enabling edge nodes to train local detection models and share only model parameters with a central server. However, the transition from centralised to federated intrusion detection introduces two new problems that the existing literature had not adequately solved. First, the choice of local loss function — which governs every gradient update at every client — had received almost no attention in the FL context, with virtually all prior work defaulting to standard Binary Cross-Entropy (BCE). Second, the statistical heterogeneity inherent in real IoT deployments (the non-IID problem) causes client gradients to diverge, degrading the quality of the globally aggregated model in ways that standard FedAvg cannot fully compensate for.

This thesis proposed, implemented, and rigorously evaluated a framework that addresses both problems through a principled application of Tsallis non-extensive statistics to federated loss function design. The following sections summarise the contributions, synthesise the findings across five benchmark datasets, reflect on the limitations of the work, and chart a course for future investigation.

6.2 Summary of Contributions

This thesis makes four distinct contributions to the field of privacy-preserving IoT security, spanning theoretical, methodological, and empirical dimensions.

6.2.1 Contribution 1: q-Sigmoid Loss for Federated IoT Intrusion Detection

The first contribution is the systematic application of the Tsallis q-Sigmoid activation function and q-Cross Entropy loss to federated intrusion detection. The q-Sigmoid generalises the standard sigmoid via the q -exponential, a power-law function that produces heavier gradient tails than the exponential function underlying BCE. This property is theoretically desirable for IoT network traffic, whose attack distributions are known to be heavy-tailed and non-Gaussian.

Formally, the q-sigmoid activation is defined as:

$$\sigma_q(z) = \frac{1}{1 + \exp_q(-z)}, \quad \exp_q(x) = [1 + (1 - q)x]_+^{1/(1-q)} \quad (6.1)$$

and the corresponding q-Cross Entropy loss is:

$$\mathcal{L}_q(y, p) = -[y \cdot \ln_q(p) + (1 - y) \cdot \ln_q(1 - p)] \quad (6.2)$$

where $\ln_q(x) = (x^{1-q} - 1)/(1 - q)$ and both functions recover BCE exactly at $q = 1$. The contribution demonstrates that a moderate departure from BCE ($q^* \approx 1.2$ – 1.4) consistently improves PR-AUC and ROC-AUC across diverse IoT security datasets without requiring any changes to the federated aggregation protocol.

6.2.2 Contribution 2: Grid Search with Strict Data Separation

The second contribution is a rigorous hyperparameter selection protocol for the entropic index q . Prior work on Tsallis-based loss functions had relied on fixed q values selected through intuition or informal comparison, which constitutes implicit data leakage when the same data is used for both selection and evaluation. This thesis establishes a clean three-way data partition (training shard, validation set, test set) and a Level-1 offline grid search in which q is selected exclusively on the validation set and all final metrics are computed on the held-out test set. This protocol is more rigorous than most prior work in the FL intrusion detection literature and provides a replicable template for future studies.

6.2.3 Contribution 3: PerClient-q-Sigmoid

The third and most Important contribution is the PerClient-q-Sigmoid mechanism, in which each federated client independently assigns its own entropic index q_k based on its

local label distribution:

$$q_k = q_{\min} + (q_{\max} - q_{\min}) \cdot \sqrt{2|\rho_k - 0.5|} \quad (6.3)$$

where ρ_k is the local attack rate and the square-root compression prevents extreme clients from receiving q_k values so large that their gradients dominate FedAvg aggregation. Each client’s learning rate is additionally scaled as $\eta_k = \eta/q_k$ to equalise the magnitude of weight updates across heterogeneous clients.

The key design property of this contribution is that the server performs *standard, unmodified FedAvg*: it aggregates weight parameters without any knowledge of the q_k values used by individual clients. This makes the method a plug-and-play enhancement that can be added to any existing FedAvg system by modifying only the client-side training loop. No additional communication, no additional server computation, and no coordination between clients is required.

6.2.4 Contribution 4: Classification Threshold Calibration for Imbalanced FL

The fourth contribution is the systematic integration of validation-set threshold tuning into the FL training loop. This thesis demonstrates that the default threshold of $\tau = 0.5$ is consistently suboptimal for imbalanced IoT intrusion data: the optimal threshold converges to values in the range $[0.38, 0.45]$ for datasets with majority-attack distributions (UNSW-NB15, WUSTLIH2021) and shifts correspondingly for minority-attack datasets (CICIDS-2018). The contribution establishes threshold calibration as a necessary post-aggregation step after every federated round, not merely a post-training fine-tuning choice, and applies it uniformly to all baseline methods to ensure fair comparison.

6.3 Summary of Experimental Findings

The framework was evaluated across three public IoT security datasets under both IID and non-IID (Dirichlet $\alpha = 0.5, 0.3, 0.1$) federated partitioning, with five independent repeated runs per setting. The key findings are summarised below.

6.3.1 Finding 1: q-Sigmoid Consistently Outperforms BCE

Across all three datasets and both partitioning strategies, the q-Sigmoid loss with grid-searched q^* achieves higher PR-AUC than BCE. The average improvement is to be determined in PR-AUC and to be determined in ROC-AUC across datasets under non-IID

partitioning. This confirms the central hypothesis that Tsallis non-extensivity is beneficial for IoT intrusion detection, where attack traffic follows heavy-tailed distributions that deviate from the sub-Gaussian assumption underlying BCE.

6.3.2 Finding 2: The Optimal q is Stable and Transferable

Grid search consistently identifies $q^* \in [1.2, 1.4]$ for four of five datasets (UNSW-NB15, WUSTL-IIoT-2021, CICIDS2018, with extreme imbalance ($> 95\%$ attacks) pushing q^* to *to be determined*). The stability of q^* across IID and non-IID settings within each dataset demonstrates that the optimal Tsallis index is a property of the data distribution rather than the partitioning configuration. This finding has practical significance: for new IoT deployments without access to labelled data for grid search, $q^* = 1.2$ is a reasonable default that performs well across diverse network environments.

6.3.3 Finding 3: PerClient-q-Sigmoid Excels Under Non-IID Conditions

The PerClient-q-Sigmoid mechanism achieves the highest PR-AUC in the non-IID setting on *to be determined* datasets and the highest ROC-AUC on *to be determined* datasets. The performance advantage over grid-tuned q-Sigmoid (the next-best method) is proportional to the inter-client variance in attack rates: on UNSW-NB15 ($\Delta\rho_{\max} = 60.6\%$ under non-IID) the advantage is to be determined PR-AUC, while on WUSTLIIOT2021 where all clients have similar high attack rates the advantage narrows to to be determined. This confirms the central design hypothesis: distribution-aware q_k assignment provides the most benefit precisely in the heterogeneous conditions it was designed for.

6.3.4 Finding 4: Threshold Tuning is Universally Critical

Without threshold tuning, the optimal threshold $\tau^* = 0.5$ would inflate recall at the expense of precision for all methods on majority-attack datasets, producing F1 scores to be determined lower than the tuned results. The magnitude of threshold deviation from 0.5 correlates with training attack rate, making threshold calibration more critical for UNSW ($\tau^* \approx 0.16\text{--}0.45$) than for CICIDS-2018 ($\tau^* \approx 0.90\text{--}0.75$). The systematic application of threshold tuning to all methods in this thesis is a methodological improvement over prior work that reports only fixed-threshold metrics.

6.3.5 Finding 5: No Single Baseline Dominates Across All Settings

GCE achieves the highest F1-Score in some configurations due to its higher recall, while Focal Loss leads on PR-AUC in others due to its conservative positive-class prediction at high thresholds ($\tau^* \approx 0.74$). PerClient-q-Sigmoid provides the most consistent performance across all three metrics, three datasets, and two partitioning strategies, making it the recommended choice for practical federated IoT security deployments where a single model must generalise across diverse network environments.

6.4 Limitations

This thesis makes significant contributions but is also subject to several limitations that should be acknowledged.

6.4.1 Linear Classifier Scope

The experimental framework uses logistic regression as the base classifier, which is a linear model. While logistic regression is interpretable, computationally lightweight (appropriate for resource-constrained IoT edge nodes), and well-suited to establishing controlled comparisons between loss functions, it cannot capture the non-linear feature interactions present in complex attack patterns. The findings of this thesis — that q-Sigmoid and PerClient-q-Sigmoid outperform BCE — are specific to linear classifiers, and the magnitude of improvement may differ when deep neural networks are used as the local model architecture.

6.4.2 Binary Classification Only

The framework performs binary classification (normal vs. attack), discarding the multi-class attack type information present in all five datasets. While this is consistent with operational intrusion detection systems that raise a binary alert, a multi-class extension would provide finer-grained threat characterisation. The extension of q-Sigmoid to categorical cross-entropy (multi-class setting) is theoretically straightforward but was not evaluated in this work.

6.4.3 Fixed Non-IID Severity

The primary non-IID experiments use a single Dirichlet concentration parameter $\alpha = 0.5$. While this represents a practically relevant degree of heterogeneity, the framework was not systematically evaluated across the full range $\alpha \in \{0.05, 0.1, 0.3, 0.5, 1.0, 5.0\}$.

Preliminary experiments suggest the PerClient advantage increases for $\alpha < 0.3$ (more extreme non-IID), but this was not confirmed across all five datasets with repeated runs.

6.4.4 Static Per-Client q Assignment

The q_k value for each client is computed once from the initial data distribution and held fixed throughout all training rounds. In real IoT deployments, the local data distribution may shift over time (concept drift) as new attack types emerge or traffic patterns change. A static q_k computed at initialisation may become suboptimal as the distribution evolves, potentially reducing the method’s effectiveness in long-running deployments.

6.4.5 Privacy Mechanism Absence

While Federated Learning inherently protects raw data by sharing only model parameters, it does not provide formal privacy guarantees. Recent work has shown that model parameters can be exploited through gradient inversion attacks to partially reconstruct training data [41]. This thesis does not integrate formal privacy mechanisms such as Differential Privacy (DP) or Secure Multi-Party Computation (SMPC), and the interaction between such mechanisms and the q -Sigmoid gradient properties was not studied.

6.5 Future Work

The contributions and limitations of this thesis open several promising directions for future research, ranging from immediate methodological extensions to longer-term system-level investigations.

6.5.1 Deep Learning Backbone with q -Sigmoid Loss

The most immediate extension is to replace logistic regression with a deep neural network — such as a multilayer perceptron (MLP), 1D convolutional network (CNN), or Transformer-based architecture — as the local model in each federated client. The q -Sigmoid loss and PerClient- q mechanism are architecture-agnostic: they modify only the final layer’s activation function and the loss computation, and can be applied to any neural network without modification.

Of particular interest is whether the PerClient- q advantage is amplified or diminished when the model has more parameters. With deeper models, client drift under non-IID conditions is more severe [42], and the distribution-aware gradient regularisation provided by heterogeneous q_k values may become more important. Conversely, deeper models may implicitly learn to handle distribution shift through representation learning, reducing the marginal value of the PerClient- q mechanism.

6.5.2 Dynamic Per-Client q : Adaptive Drift Detection

A natural extension of the static PerClient- q mechanism is to allow q_k to update dynamically as the local data distribution evolves. This is particularly relevant in IoT environments subject to concept drift, where the prevalence of specific attack types changes over time. The proposed adaptive mechanism would:

1. Monitor the local attack rate $\rho_k^{(t)}$ at each round t .
2. Detect significant distribution shift using a statistical test (e.g. Page-Hinkley test or ADWIN).
3. Recompute q_k from the updated $\rho_k^{(t)}$ when drift is detected.
4. Apply a smoothing factor to prevent abrupt changes: $q_k^{(t)} = \beta \cdot q_k^{(t-1)} + (1 - \beta) \cdot q_k^{\text{new}}$.

This would make the framework responsive to temporal changes in IoT network environments, addressing the static assignment limitation identified in Section 6.4.

6.5.3 Differential Privacy Integration

Formal privacy guarantees are increasingly required in IoT deployments subject to data protection regulations (e.g. GDPR, CCPA). Differentially Private FL (DP-FL) [43] adds calibrated Gaussian noise to client gradient updates before transmission to the server, bounding the information leakage about any individual training sample. A critical open question is how the q -Sigmoid gradient properties interact with DP noise: specifically, whether the heavier gradient tails of the q -Sigmoid loss make the model more or less sensitive to gradient perturbation.

The proposed investigation would vary the privacy budget $\varepsilon \in \{0.1, 0.5, 1.0, 5.0, 10.0\}$ and measure the PR-AUC degradation for BCE versus q -Sigmoid under both IID and non-IID conditions. If the q -Sigmoid gradient’s power-law tail is more robust to Gaussian noise than BCE’s exponential tail (a plausible hypothesis given the different curvature properties), this would provide an additional practical motivation for the q -Sigmoid approach in privacy-sensitive deployments.

6.5.4 Multi-Class q -Cross Entropy for Fine-Grained Threat Detection

The binary classification framework of this thesis discards attack type information. A multi-class extension using the Tsallis q -Cross Entropy in its categorical form would enable detection and classification of specific attack categories (DoS, Reconnaissance, Data Exfiltration, etc.) simultaneously.

The categorical q-Cross Entropy is defined as:

$$\mathcal{L}_q^{\text{multi}}(y, \mathbf{p}) = - \sum_{c=1}^C y_c \cdot \ln_q(p_c) \quad (6.4)$$

where C is the number of classes and $\mathbf{p} \in \Delta^{C-1}$ is the softmax probability vector. The PerClient-q mechanism would extend naturally to this setting by computing a per-client q_k from the local class distribution entropy rather than a binary imbalance measure. Clients with highly skewed class distributions (e.g. a gateway that primarily sees DoS traffic) would receive higher q_k , while clients with more uniform attack-type distributions would receive q_k closer to 1.

6.5.5 q-Sigmoid with Personalised Federated Learning

Personalised Federated Learning (pFL) [44] trains a global model that is subsequently fine-tuned to each client’s local distribution, producing client-specific models rather than a single global model. The interaction between pFL fine-tuning and the q-Sigmoid loss is an unexplored area with practical significance: fine-tuning a globally trained q-Sigmoid model with a client-specific q_k value may produce detection models better calibrated to each IoT gateway’s specific traffic profile than either global FL or local-only training.

This direction connects the contributions of this thesis to the broader pFL literature and could provide a principled theoretical framework for understanding when PerClient-q-Sigmoid is equivalent to, or a special case of, personalised federated learning.

6.5.6 Communication-Efficient q-Sigmoid FL

Federated Learning in resource-constrained IoT environments must minimise communication overhead. Two directions are relevant here. First, the combination of q-Sigmoid training with gradient compression or quantisation (e.g. TopK sparsification, 1-bit quantisation) could reduce the per-round communication cost while preserving the loss function’s beneficial properties. Second, asynchronous FL protocols — where clients communicate with the server at irregular intervals rather than in synchronised rounds — are more realistic for IoT environments with intermittent connectivity, and the stability of the PerClient-q mechanism under asynchronous aggregation warrants investigation.

6.5.7 Real-World Deployment Study

All experiments in this thesis were conducted in simulation using offline datasets. A real-world deployment study on physical IoT hardware (e.g. Raspberry Pi gateway nodes, industrial Modbus controllers) would validate the practical feasibility of the framework. Specific questions of interest include:

- **Computational overhead:** Does the q-Sigmoid activation’s power function $x \mapsto x^{1/(1-q)}$ introduce significant latency compared to the standard exponential on embedded hardware?
- **Memory footprint:** The logistic regression model used in this thesis has a small parameter count (~ 43 floats for UNSW-NB15), but deep learning extensions may require model compression techniques (pruning, knowledge distillation) for deployment on microcontrollers.
- **Live attack detection:** Evaluation on live network traffic with real adversarial actors would test the framework’s generalisation to attack variations not present in the offline benchmark datasets.

6.6 Broader Impact

The work presented in this thesis has implications beyond the specific problem of IoT intrusion detection. The core insight — that the Tsallis entropic index can be matched to local data characteristics within a federated system, improving global model quality without server-side modification — is general and applies to any binary or multi-class classification problem in federated settings where clients have heterogeneous class distributions.

Potential application domains beyond IoT security include federated fraud detection in distributed financial networks (where merchant nodes have very different fraud rates), federated clinical decision support (where hospitals have different patient demographics and disease prevalences), and federated content moderation (where geographic nodes see different proportions of policy-violating content). In each of these domains, the non-IID problem is pervasive, class imbalance is severe, and privacy constraints preclude data centralisation — precisely the conditions under which this thesis demonstrates the greatest advantage for q-Sigmoid and PerClient-q-Sigmoid.

From a data privacy perspective, the framework’s compatibility with existing FedAvg infrastructure means that privacy guarantees provided by the underlying FL system (parameter-only sharing, optional differential privacy, secure aggregation) are fully preserved. The q-Sigmoid modification does not require any additional information to be shared beyond what standard FedAvg transmits, ensuring that the privacy-utility trade-off of the underlying FL system is not adversely affected.

Bibliography

- [1] IoT Analytics, “State of iot 2025: Number of connected iot devices,” 2025, industry report. [Online]. Available: <https://iot-analytics.com>
- [2] Statista, “Internet of things (iot) adoption and industry trends,” 2025, statistical report. [Online]. Available: <https://www.statista.com>
- [3] A. H. Lashkari *et al.*, “Machine learning-based intrusion detection methods in iot,” *Electronics*, vol. 13, no. 18, p. 3601, 2024.
- [4] P. Gugueoth, R. Shukla, and K. Rao, “Security of internet of things (iot) using federated learning,” *Journal of Network and Computer Applications*, vol. 214, p. 103546, 2023.
- [5] M. Aziz Al Kabir, W. Elmedany, and M. S. Sharif, “Securing iot devices against emerging security threats: Challenges and mitigation techniques,” *Journal of Cyber Security Technology*, vol. 7, no. 4, pp. 199–223, 2023.
- [6] S. T. Erukude, V. C. Marella, and S. R. Veluru, “Ai-driven cybersecurity threats: A survey of emerging risks and defensive strategies,” in *Data Science and Applications*, ser. Lecture Notes in Networks and Systems. Cham: Springer, 2026, vol. 1723.
- [7] S. Dritsas, P. Konstantinou, A. Fragkiadakis, and K. Fysarakis, “Federated learning for iot: A survey of techniques, challenges, and applications,” *Sensors*, vol. 14, no. 1, p. 9, 2025.
- [8] R. L. Villanueva *et al.*, “Iot anomaly detection based on real-time traffic monitoring,” *Future Internet*, vol. 16, no. 5, p. 194, 2024.
- [9] S. Garcia, A. Parmisano, and M. J. Erquiaga, “Iot-23: A labeled dataset with malicious and benign iot network traffic,” 2020, [Data set]. [Online]. Available: <http://doi.org/10.5281/zenodo.4743746>
- [10] N. Pandey and P. K. Mishra, “Performance analysis of entropy variation-based detection of ddos attacks in iot,” *Internet of Things*, vol. 23, p. 100812, 2023.
- [11] E. M. Campos, P. F. Saura, A. González-Vidal, J. L. Hernández-Ramos, J. B. Bernabé, G. Baldini, and A. Skarmeta, “Evaluating federated learning for intrusion detection in internet of things: Review and challenges,” *Computer Networks*, vol. 203, p. 108661, 2022.

- [12] L. Atzori, A. Iera, and G. Morabito, “The internet of things: A survey,” *Computer Networks*, vol. 54, no. 15, pp. 2787–2805, 2010.
- [13] J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami, “Internet of things (iot): A vision, architectural elements, and future directions,” *Future Generation Computer Systems*, vol. 29, no. 7, pp. 1645–1660, 2013.
- [14] H. F. Pardede, P. Adhi, V. Zilvan, A. R. Yuliani, and Others, “A generalization of sigmoid loss function using tsallis statistics for binary classification,” *Entropy*, vol. 24, no. 11, p. 1576, 2022.
- [15] M. Aziz Al Kabir *et al.*, “Ai-driven cybersecurity threats: Emerging risks and defensive strategies,” *Journal of Cyber Security Technology*, vol. 7, no. 4, pp. 199–223, 2023.
- [16] A. H. Lashkari *et al.*, “Machine learning-based intrusion detection methods for iot security,” *Electronics*, vol. 13, no. 18, p. 3601, 2024.
- [17] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y. Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*. PMLR, 2017, pp. 1273–1282.
- [18] E. Bagdasaryan *et al.*, “How to backdoor federated learning,” in *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*. PMLR, 2020, pp. 2938–2948.
- [19] K. Pillutla *et al.*, “Robust aggregation for federated learning,” *IEEE Transactions on Signal Processing*, vol. 70, pp. 1142–1154, 2022.
- [20] C. Tsallis, “Possible generalization of boltzmann-gibbs statistics,” *Journal of Statistical Physics*, vol. 52, no. 1-2, pp. 479–487, 1988.
- [21] H. F. Pardede *et al.*, “A generalization of sigmoid loss function using tsallis statistics for binary classification,” *Entropy*, vol. 24, no. 11, p. 1576, 2022.
- [22] S. A. Rahman *et al.*, “Internet of things intrusion detection: Centralized, on-device, or federated learning?” *IEEE Network*, vol. 34, no. 6, pp. 310–317, 2020.
- [23] T. Zhang *et al.*, “Federated learning for internet of things: Applications, challenges, and opportunities,” *IEEE Internet of Things Journal*, vol. 9, no. 11, pp. 10 285–10 304, 2022.
- [24] R. L. Villanueva *et al.*, “Iot anomaly detection based on real-time traffic monitoring,” *Future Internet*, vol. 16, no. 5, p. 194, 2024.

- [25] A. Khraisat *et al.*, “Federated learning for intrusion detection in iot environments,” *Cluster Computing*, 2025.
- [26] P. Gugueoth and R. Sharma, “Security of internet of things using federated learning,” *Journal of Network and Computer Applications*, vol. 214, p. 103546, 2023.
- [27] C. M. Bishop, *Pattern Recognition and Machine Learning*. Springer, 2006.
- [28] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.
- [29] Z. Zhang and M. R. Sabuncu, “Generalized cross entropy loss for training deep neural networks with noisy labels,” *Advances in Neural Information Processing Systems*, vol. 31, pp. 8778–8788, 2018.
- [30] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 2980–2988.
- [31] X. Li, K. Huang, W. Yang, S. Wang, and Z. Zhang, “On the convergence of fedavg on non-iid data,” *International Conference on Learning Representations (ICLR)*, 2020.
- [32] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, “Communication-efficient learning of deep networks from decentralized data,” in *Artificial Intelligence and Statistics (AISTATS)*, 2017.
- [33] Z. Zhang and M. Sabuncu, “Generalized cross entropy loss for training deep neural networks with noisy labels,” in *Advances in Neural Information Processing Systems*, vol. 31, 2018.
- [34] N. Moustafa and J. Slay, “Unsw-nb15: A comprehensive data set for network intrusion detection systems,” in *Military Communications and Information Systems Conference (MilCIS)*. IEEE, 2015, pp. 1–6.
- [35] M. Zolanvari, M. A. Teixeira, L. Gupta, K. M. Khan, and R. Jain, “Machine learning-based network vulnerability analysis of industrial internet of things,” *IEEE Internet of Things Journal*, vol. 6, no. 4, pp. 6822–6834, 2019.
- [36] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, “Toward generating a new intrusion detection dataset and intrusion traffic characterization,” in *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP)*. SciTePress, 2018, pp. 108–116.
- [37] C. Tsallis, “Possible generalization of boltzmann-gibbs statistics,” *Journal of Statistical Physics*, vol. 52, no. 1-2, pp. 479–487, 1988.

- [38] K. Hsieh, A. Phanishayee, O. Mutlu, and P. Gibbons, “Quagmire: Non-iid data in federated learning,” in *International Conference on Machine Learning*, 2020.
- [39] Q. Li, Y. Diao, Q. Chen, and B. He, “Federated learning on non-iid data silos: An experimental study,” *IEEE International Conference on Data Engineering*, 2022.
- [40] J. Davis and M. Goadrich, “The relationship between precision-recall and roc curves,” in *Proceedings of the 23rd International Conference on Machine Learning*, 2006, pp. 233–240.
- [41] J. Geiping, H. Bauermeister, H. Dröge, and M. Moeller, “Inverting gradients – how easy is it to break privacy in federated learning?” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 16 937–16 947.
- [42] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. T. Suresh, “Scaffold: Stochastic controlled averaging for federated learning,” in *International Conference on Machine Learning*. PMLR, 2020, pp. 5132–5143.
- [43] H. B. McMahan, D. Ramage, K. Talwar, and L. Zhang, “Learning differentially private recurrent language models,” in *International Conference on Learning Representations*, 2018.
- [44] A. Fallah, A. Mokhtari, and A. Ozdaglar, “Personalized federated learning with theoretical guarantees: A model-agnostic meta-learning approach,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 3557–3568.